

AN ELEMENTARY  
INTRODUCTION TO  
MICROECONOMICS



KUTUSH KUMAR, KUTUM PAKHI

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MICROECONOMICS



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## TO THE READER

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This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.<sup>1</sup>

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

---

## HOW A CHAPTER RUNS

---

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

*This book teaches An Elementary Introduction to Microeconomics one idea at a time, from the very beginning.*

*The narrow column beside the text holds short notes.*

**Key term.** A new word, shown in bold where it first appears, with its meaning in one line.

*A bold word is a new term, and its meaning waits beside it.*

*Some everyday words have a second, special meaning in this subject.*

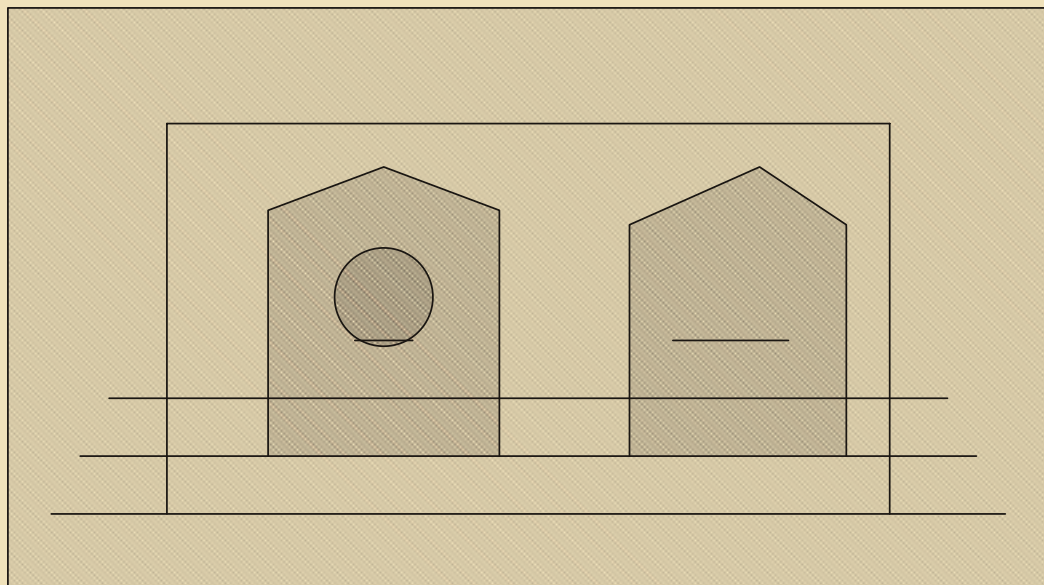
*Each chapter has the same shape, so you always know where you are.*

*Chapters refer to one another by name, never by page.*

---

<sup>1</sup> A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.





*a merchant's counting-house with coin chests, scales and a clerk at a high desk, 1860s*

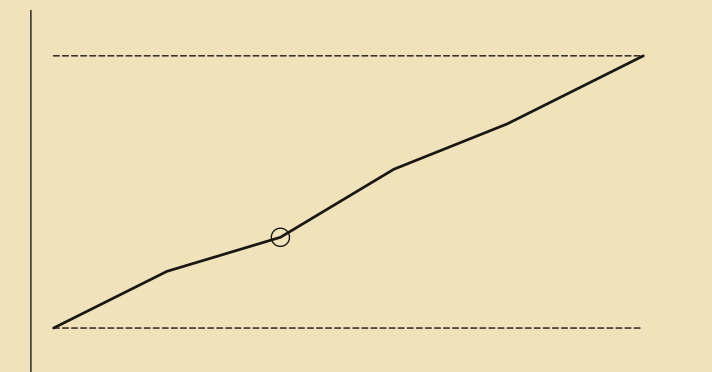
## CHAPTER I

# What Finance Is

*A practical question about what finance is*



*This chapter introduces what finance is for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 1.1. *What Finance Is figure 1*

*This section builds the idea of finance in plain words before moving to application.*

**Finance.** The activity of finding money, using it, and judging whether the result was worth it.

*The term finance gives the chapter a precise name for this part of the problem.*

**Cash flow.** Money that actually moves into or out of a business.

*The term cash flow gives the chapter a precise name for this part of the problem.*

**Profit.** What a business earns from sales after subtracting the costs of earning them.

*The term profit gives the chapter a precise name for this part of the problem.*

*The central question in finance in plain words is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

## 1.1 FINANCE IN PLAIN WORDS

The purpose here is to make finance in plain words concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Finance** means The activity of finding money, using it, and judging whether the result was worth it. A shop owner who borrows to buy stock does finance. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash flow** means Money that actually moves into or out of a business. Paying a supplier Tk 5,000 is a cash flow of minus Tk 5,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Profit** means What a business earns from sales after subtracting the costs of earning them. Sales of Tk 100 and costs of Tk 80 leave profit of Tk 20. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For finance in plain words, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

## 1.2 FINANCE BESIDE ITS NEIGHBOURING SUBJECTS

The purpose here is to make finance beside its neighbouring subjects concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating efficiency** means How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sole proprietorship** means A business owned and run by one person, who keeps all profit and answers for all debts. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Partnership** means A business owned by two or more people who agree to run it together and share its profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For finance beside its neighbouring subjects, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of finance beside its neighbouring subjects before moving to application.*

**Operating efficiency.** How much useful output a business gets from the money and effort it uses.

*The term operating efficiency gives the chapter a precise name for this part of the problem.*

**Sole proprietorship.** A business owned and run by one person, who keeps all profit and answers for all debts..

*The term sole proprietorship gives the chapter a precise name for this part of the problem.*

**Partnership.** A business owned by two or more people who agree to run it together and share its profit..

*The term partnership gives the chapter a precise name for this part of the problem.*

*The central question in finance beside its neighbouring subjects is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

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### 1.3 THE FINANCE FUNCTION INSIDE A FIRM

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*This section builds the idea of the finance function inside a firm before moving to application.*

**Company.** A business set up as a legal person of its own, owned through shares, so the owners are separate from the business..

*The term company gives the chapter a precise name for this part of the problem.*

*The central question in the finance function inside a firm is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

The purpose here is to make the finance function inside a firm concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Company** means A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the finance function inside a firm, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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#### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to what finance is. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

*A short restatement keeps the financial meaning attached to the calculation.*

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of What Finance Is is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

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## WHERE PEOPLE GO WRONG

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Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

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## IN HISTORY

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The Dutch East India Company (VOC), chartered 1602, sold shares to the public that traded in Amsterdam, an early company separating owners from managers. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

*A historical case shows how the chapter's financial problem appeared in practice.*

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**REVIEW QUESTIONS**

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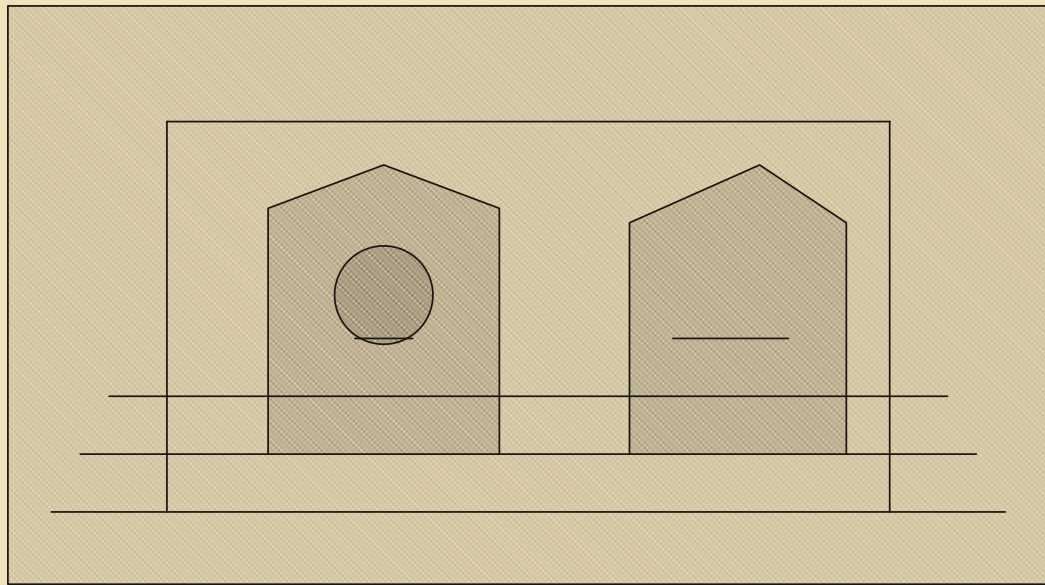
**01.T1** say in plain words what finance covers and how it differs from accounting and from management

**01.T2** tell profit from cash flow using a new set of sales and collection figures

**01.T3** compare sole proprietorship, partnership and company for a new tailoring shop

**PAUSE AND CHECK 01.P1** After the finance definition, state the rule in your own words.

**PAUSE AND CHECK 01.P2** After profit versus cash flow, state the rule in your own words.



*a banker at a high desk reviewing a ledger by gaslight, 1880s*

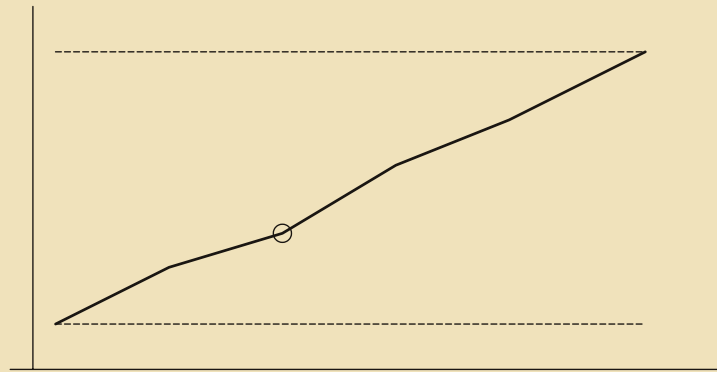
## CHAPTER II

# The Financial Manager's Decisions

*A practical question about the financial manager's decisions*



*This chapter introduces the financial manager's decisions for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 2.1. *The Financial Manager's Decisions figure 1*

*This section builds the idea of the three big decisions before moving to application.*

**Financial manager.** The person who decides how a business finds, uses and protects its money.

*The term financial manager gives the chapter a precise name for this part of the problem.*

**Investment decision.** A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money..

*The term investment decision gives the chapter a precise name for this part of the problem.*

**Financing decision.** A choice about where the money comes from: owners' own funds, a loan, or new owners..

*The term financing decision gives the chapter a precise name for this part of the problem.*

**Dividend.** A share of a company's profit paid out to its owners instead of being kept in the business..

*The term dividend gives the chapter a precise name for this part of the problem.*

*The central question in the three big decisions is what changes when the underlying decision changes.*

## 2.1 THE THREE BIG DECISIONS

The purpose here is to make the three big decisions concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial manager** means The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Investment decision** means A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financing decision** means A choice about where the money comes from: owners' own funds, a loan, or new owners. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Dividend** means A share of a company's profit paid out to its owners instead of being kept in the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the three big decisions, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

## 2.2 WHAT A FINANCIAL MANAGER DOES

The purpose here is to make what a financial manager does concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend decision** means A choice about how much profit to pay out to owners and how much to keep for the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Treasurer** means The officer in charge of handling cash, raising money and keeping relations with banks. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Controller** means The officer in charge of accounting records, reports and checking that money is used as planned. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial planning** means Estimating the money a business will need and receive in future, and arranging how to meet the gap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of what a financial manager does before moving to application.*

**Dividend decision.** A choice about how much profit to pay out to owners and how much to keep for the business..

*The term dividend decision gives the chapter a precise name for this part of the problem.*

**Treasurer.** The officer in charge of handling cash, raising money and keeping relations with banks..

*The term treasurer gives the chapter a precise name for this part of the problem.*

**Controller.** The officer in charge of accounting records, reports and checking that money is used as planned..

*The term controller gives the chapter a precise name for this part of the problem.*

**Financial planning.** Estimating the money a business will need and receive in future, and arranging how to meet the gap..

*The term financial planning gives the chapter a precise name for this part of the problem.*

*The central question in what a financial manager does is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of decisions on the balance sheet before moving to application.*

**Balance sheet.** A statement listing what a business owns and what it owes on one date..

*The term balance sheet gives the chapter a precise name for this part of the problem.*

**Asset.** Something a business owns that has value, such as cash, stock or a van..

*The term asset gives the chapter a precise name for this part of the problem.*

**Liability.** Money a business owes to others, such as a bank loan or an unpaid supplier bill..

*The term liability gives the chapter a precise name for this part of the problem.*

**Equity.** What is left for the owners when liabilities are subtracted from assets..

*The term equity gives the chapter a precise name for this part of the problem.*

*The central question in decisions on the balance sheet is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what a financial manager does, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## 2.3 DECISIONS ON THE BALANCE SHEET

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The purpose here is to make decisions on the balance sheet concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Balance sheet** means A statement listing what a business owns and what it owes on one date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Asset** means Something a business owns that has value, such as cash, stock or a van. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Liability** means Money a business owes to others, such as a bank loan or an unpaid supplier bill. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Equity** means What is left for the owners when liabilities are subtracted from assets. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For decisions on the balance sheet, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a

later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

### WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the financial manager's decisions. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*



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## SUMMARY

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The central purpose of The Financial Manager's Decisions is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

---

## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

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## IN HISTORY

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General Motors in the 1920s: Donaldson Brown, who had come from DuPont, built financial controls linking return on investment to decisions. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

*A historical case shows how the chapter's financial problem appeared in practice.*

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## REVIEW QUESTIONS

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**02.T1** name and tell apart the three decisions using a new business

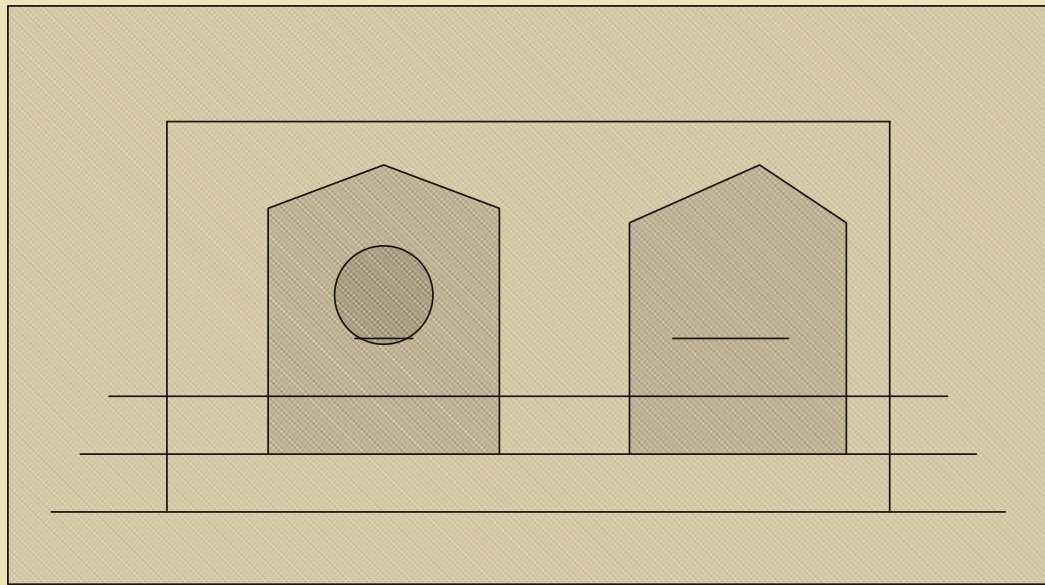
**02.T2** state the work of a treasurer and a controller

**02.T3** build a balance sheet from a new list of items and check that it balances

**PAUSE AND CHECK 02.P1** After the three decisions, state the rule in your own words.

**PAUSE AND CHECK 02.P2** After the balance sheet, state the rule in your own words.





*a company boardroom table with directors, quill pens and a bound minute book, 1870s*

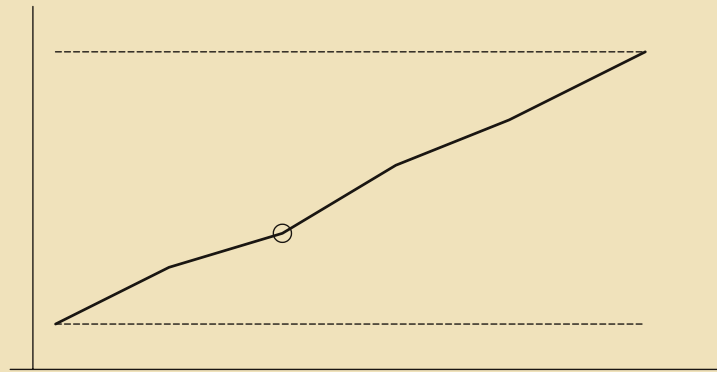
### CHAPTER III

# The Goal of the Firm and Agency Theory

*A practical question about the goal of the firm and agency theory*



*This chapter introduces the goal of the firm and agency theory for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 3.1. *The Goal of the Firm and Agency Theory figure 1*

*This section builds the idea of profit maximisation and its limits before moving to application.*

**Profit maximisation.** Aiming to make the largest profit possible, usually within one year..

*The term profit maximisation gives the chapter a precise name for this part of the problem.*

**Wealth maximisation.** Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is..

*The term wealth maximisation gives the chapter a precise name for this part of the problem.*

**Shareholder.** A person who owns part of a company by holding its shares..

*The term shareholder gives the chapter a precise name for this part of the problem.*

**Share price.** The price at which one share of a company can be bought or sold today..

*The term share price gives the chapter a precise name for this part of the problem.*

*The central question in profit maximisation and its limits is what changes when the underlying decision changes.*

### 3.1 PROFIT MAXIMISATION AND ITS LIMITS

The purpose here is to make profit maximisation and its limits concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profit maximisation** means Aiming to make the largest profit possible, usually within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Wealth maximisation** means Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Shareholder** means A person who owns part of a company by holding its shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Share price** means The price at which one share of a company can be bought or sold today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For profit maximisation and its limits, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

## 3.2 WEALTH MAXIMISATION

The purpose here is to make wealth maximisation concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Stakeholder** means Anyone affected by a firm's actions, such as staff, customers, lenders and the community. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Principal** means A person who hires another to act for them. A shop owner who hires a manager is a principal. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Agent** means A person who acts for a principal and makes decisions for the principal. A hired shop manager is an agent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Agency problem** means A conflict that arises when an agent's own interests differ from those of the principal. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of wealth maximisation before moving to application.*

**Stakeholder.** Anyone affected by a firm's actions, such as staff, customers, lenders and the community..

*The term stakeholder gives the chapter a precise name for this part of the problem.*

**Principal.** A person who hires another to act for them.

*The term principal gives the chapter a precise name for this part of the problem.*

**Agent.** A person who acts for a principal and makes decisions for the principal.

*The term agent gives the chapter a precise name for this part of the problem.*

**Agency problem.** A conflict that arises when an agent's own interests differ from those of the principal..

*The term agency problem gives the chapter a precise name for this part of the problem.*

*The central question in wealth maximisation is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of owners, managers and agency theory before moving to application.*

**Agency cost.** A cost the principal bears to reduce the agency problem or from the loss it still causes..

*The term agency cost gives the chapter a precise name for this part of the problem.*

**Incentive.** A reward, such as a bonus, that encourages a person to act in a chosen way..

*The term incentive gives the chapter a precise name for this part of the problem.*

*The central question in owners, managers and agency theory is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For wealth maximisation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

### 3.3 OWNERS, MANAGERS AND AGENCY THEORY

The purpose here is to make owners, managers and agency theory concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Agency cost** means A cost the principal bears to reduce the agency problem or from the loss it still causes. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

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\* \* \*

### WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the goal of the firm and agency theory. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the

units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of The Goal of the Firm and Agency Theory is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

---

## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

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## IN HISTORY

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Berle and Means, The Modern Corporation and Private Property (1932), and Jensen and Meckling (1976) on agency costs. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

*A historical case shows how the chapter's financial problem appeared in practice.*

*The chapter points back to related ideas elsewhere in the book.*

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## SOURCES

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Chapter 1, 'What Finance Is', section 1.1 'Finance in plain words' — profit as defined there; Chapter 5, 'Present Value and Discounting', section 5.1 'Discounting a single sum' — how timing is measured in wealth..

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## REVIEW QUESTIONS

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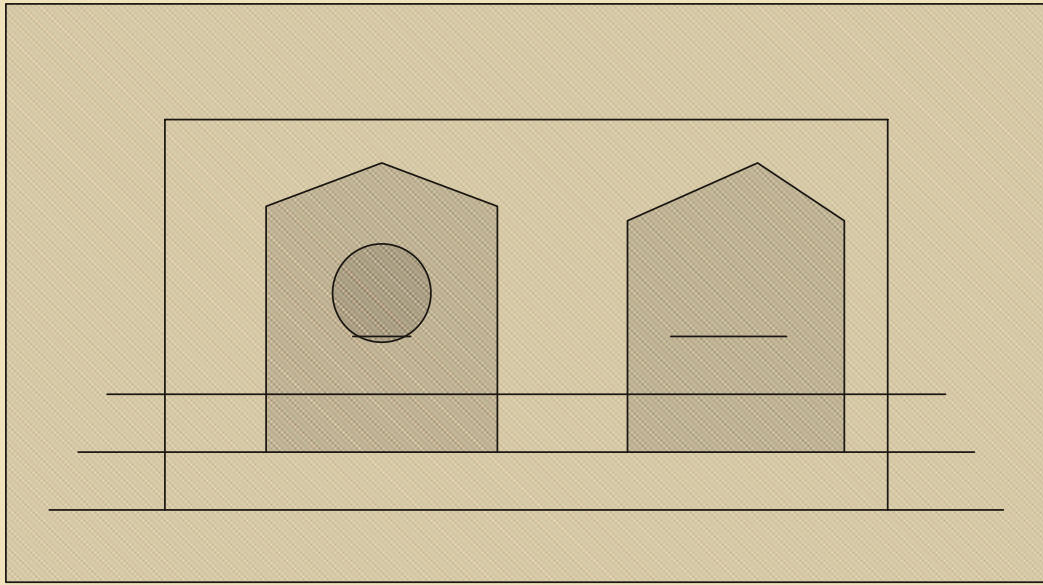
**03.T1** explain why the aim of the largest profit can mislead, using new yearly figures

**03.T2** define wealth maximisation and say how it differs from profit maximisation

**03.T3** describe an agency problem and two remedies in a hospital-equipment firm

**PAUSE AND CHECK 03.P1** After profit maximisation, state the rule in your own words.

**PAUSE AND CHECK 03.P2** After the agency relationship, state the rule in your own words.



*an hourglass, a coin purse and an open savings passbook arranged as a still life, 1850s*

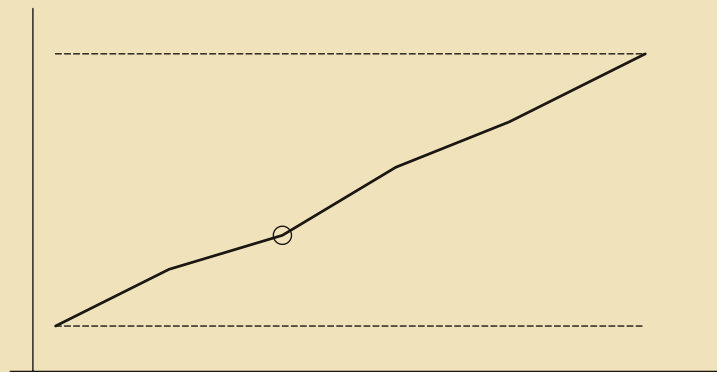
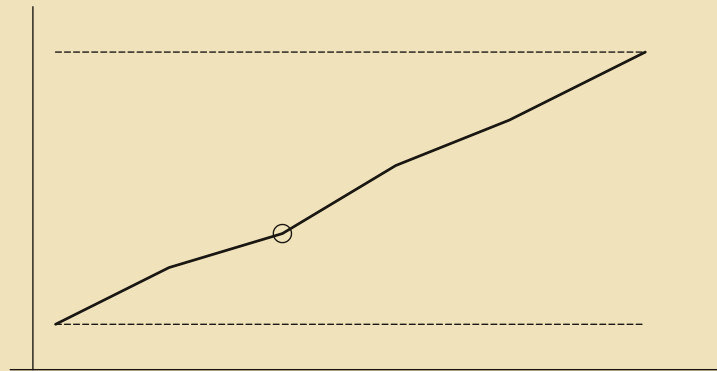
#### CHAPTER IV

## Interest and Future Value

*A practical question about interest and future value*



*This chapter introduces interest and future value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 4.1. *Interest and Future Value figure 1*FIG. 4.2. *Interest and Future Value figure 2*

*This section builds the idea of why time changes the value of money before moving to application.*

**Time value of money.** The idea that a taka received today is worth more than a taka received later..

*The term time value of money gives the chapter a precise name for this part of the problem.*

**Opportunity cost.** What you give up by choosing one use of money instead of the best alternative..

*The term opportunity cost gives the chapter a precise name for this part of the problem.*

---

#### 4.1 WHY TIME CHANGES THE VALUE OF MONEY

---

The purpose here is to make why time changes the value of money concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Time value of money** means The idea that a taka received today is worth more than a taka received later. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Opportunity cost** means What you give up by choosing one use of money instead of the best alternative. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Interest** means The payment a borrower makes to a lender for the use of money, or the reward a saver earns. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Interest rate** means The interest for one period written as a percentage of the sum, for example 8% a year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why time changes the value of money, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## 4.2 SIMPLE AND COMPOUND INTEREST

---

The purpose here is to make simple and compound interest concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Principal sum** means The original amount lent, borrowed or deposited, before any interest is added. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Simple interest** means Interest calculated only on the principal sum, so it is the same every period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

**Interest.** The payment a borrower makes to a lender for the use of money, or the reward a saver earns..

*The term interest gives the chapter a precise name for this part of the problem.*

**Interest rate.** The interest for one period written as a percentage of the sum, for example 8% a year..

*The term interest rate gives the chapter a precise name for this part of the problem.*

*The central question in why time changes the value of money is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of simple and compound interest before moving to application.*

**Principal sum.** The original amount lent, borrowed or deposited, before any interest is added..

*The term principal sum gives the chapter a precise name for this part of the problem.*

**Simple interest.** Interest calculated only on the principal sum, so it is the same every period..

*The term simple interest gives the chapter a precise name for this part of the problem.*

**Compound interest.** Interest calculated on the principal sum and also on interest already earned..

*The term compound interest gives the chapter a precise name for this part of the problem.*

**Future value.** What a sum today will be worth at a later date when it earns interest..

*The term future value gives the chapter a precise name for this part of the problem.*

*The central question in simple and compound interest is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

In this chapter, **Compound interest** means Interest calculated on the principal sum and also on interest already earned. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Future value** means What a sum today will be worth at a later date when it earns interest. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For simple and compound interest, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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### 4.3 FUTURE VALUE OF A SINGLE SUM

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*This section builds the idea of future value of a single sum before moving to application.*

**Compounding period.** The length of time after which interest is added to the balance, such as a year or six months..

*The term compounding period gives the chapter a precise name for this part of the problem.*

**Timeline.** A line marking each period, with the cash flow written at the point where it occurs..

*The term timeline gives the chapter a precise name for this part of the problem.*

The purpose here is to make future value of a single sum concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Compounding period** means The length of time after which interest is added to the balance, such as a year or six months. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Timeline** means A line marking each period, with the cash flow written at the point where it occurs. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Factor table** means A table of ready-made multipliers that saves repeating a long calculation. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For future value of a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

### WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to interest and future value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when

**Factor table.** A table of ready-made multipliers that saves repeating a long calculation..

*The term factor table gives the chapter a precise name for this part of the problem.*

*The central question in future value of a single sum is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Interest and Future Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

*Finance becomes stronger when a number is followed by a question about meaning.*

*Most errors arise from mixing terms, timing, signs, or decision rules.*

*A historical case shows how the chapter's financial problem appeared in practice.*

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### THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

---

### WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

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### IN HISTORY

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Richard Price, the 1772 Appeal on the National Debt, and the British sinking fund of 1786 that his compound-interest argument helped to inspire. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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### REVIEW QUESTIONS

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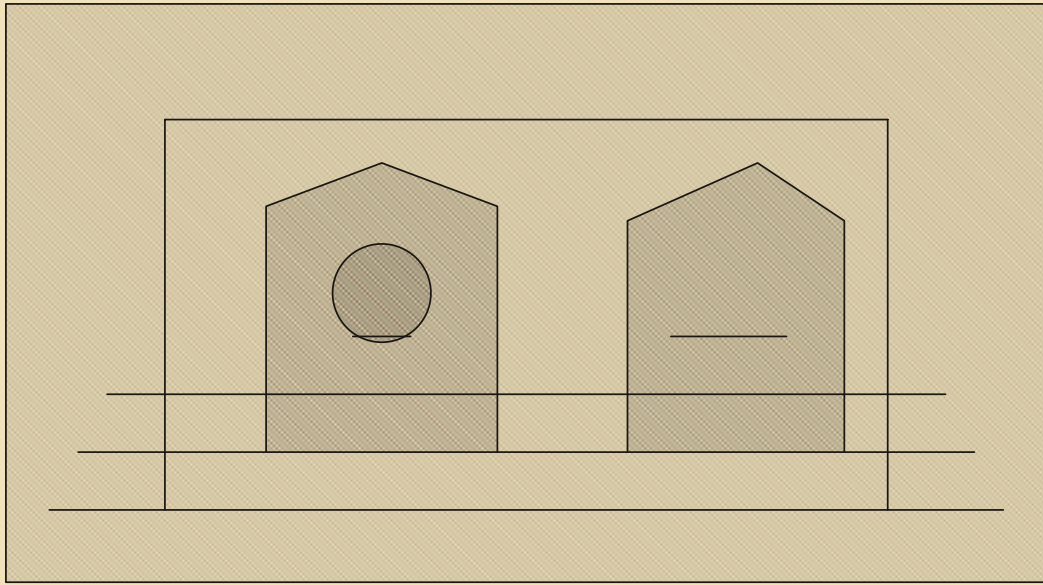
**04.T1** explain why a taka today beats a taka next year, with new figures

**04.T2** compute simple and compound interest on Tk 50,000 at 12% for four years

**04.T3** use a factor table to find a future value for a new rate and period

**PAUSE AND CHECK 04.P1** After opportunity cost, state the rule in your own words.

**PAUSE AND CHECK 04.P2** After compound interest, state the rule in your own words.



*a pocket watch and a small balance scale holding a coin against a bag of grain, still life, 1860s*

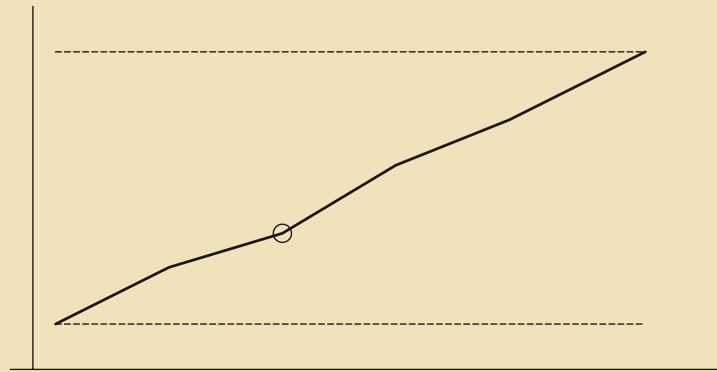
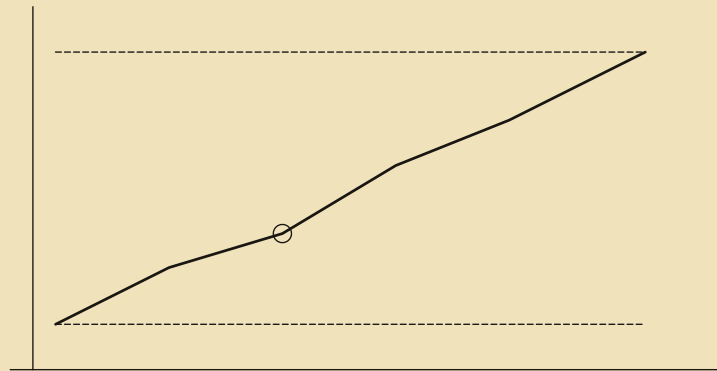
## CHAPTER V

# Present Value and Discounting

*A practical question about present value and discounting*



*This chapter introduces present value and discounting for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 5.1. *Present Value and Discounting figure 1*FIG. 5.2. *Present Value and Discounting figure 2*


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## 5.1 DISCOUNTING A SINGLE SUM

---

*This section builds the idea of discounting a single sum before moving to application.*

**Present value.** What a sum due in the future is worth today..

*The term present value gives the chapter a precise name for this part of the problem.*

**Discounting.** Finding present value by removing the interest a sum could earn before it arrives..

*The term discounting gives the chapter a precise name for this part of the problem.*

The purpose here is to make discounting a single sum concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Present value** means What a sum due in the future is worth today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discounting** means Finding present value by removing the interest a sum could earn before it arrives. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For discounting a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

## 5.2 MIXED STREAMS OF CASH FLOWS

The purpose here is to make mixed streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Discount rate** means The yearly percentage used in discounting; it states the return you could earn elsewhere. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash inflow** means Money coming into a business or to an investor, shown as a positive number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For mixed streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

*The central question in discounting a single sum is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of mixed streams of cash flows before moving to application.*

**Discount rate.** The yearly percentage used in discounting; it states the return you could earn elsewhere..

*The term discount rate gives the chapter a precise name for this part of the problem.*

**Cash inflow.** Money coming into a business or to an investor, shown as a positive number..

*The term cash inflow gives the chapter a precise name for this part of the problem.*

*The central question in mixed streams of cash flows is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

\* \* \*

---

### 5.3 COMPARING STREAMS OF CASH FLOWS

---

*This section builds the idea of comparing streams of cash flows before moving to application.*

**Cash outflow.** Money going out of a business or from an investor, shown as a negative number..

*The term cash outflow gives the chapter a precise name for this part of the problem.*

**Mixed stream.** A series of cash flows of different sizes in different periods..

*The term mixed stream gives the chapter a precise name for this part of the problem.*

*The central question in comparing streams of cash flows is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

The purpose here is to make comparing streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Cash outflow** means Money going out of a business or from an investor, shown as a negative number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Mixed stream** means A series of cash flows of different sizes in different periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For comparing streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to present value and discounting. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Present Value and Discounting is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

---

## THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

---

## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

*A historical case shows how the chapter's financial problem appeared in practice.*

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## IN HISTORY

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Irving Fisher, *The Theory of Interest* (1930), which states the discounting idea in its modern form. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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## REVIEW QUESTIONS

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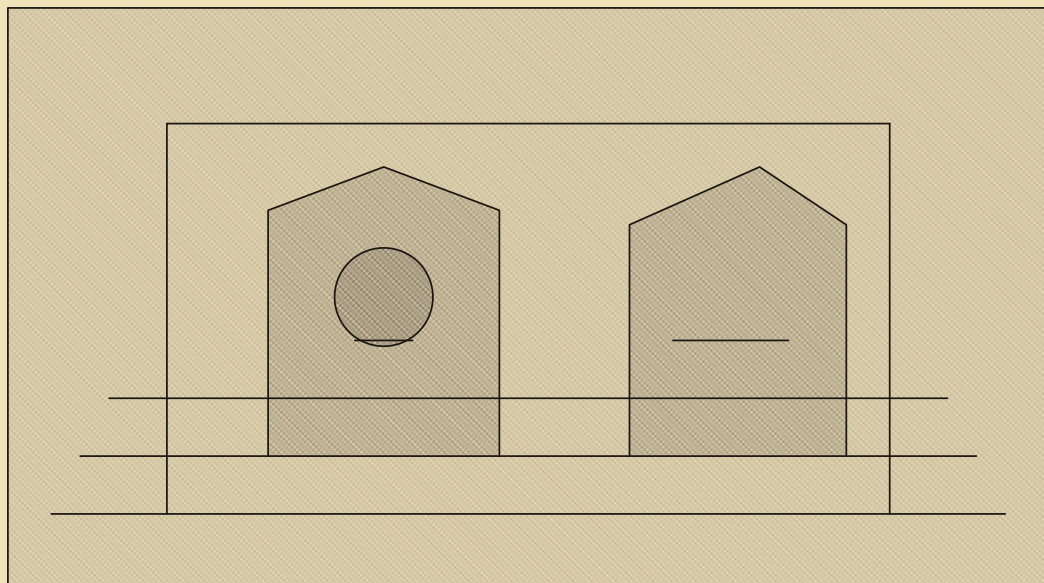
**05.T1** explain discounting as the reverse of compounding using new figures

**05.T2** find the present value of a single sum for two discount rates

**05.T3** find present value of a mixed stream and compare two streams

**PAUSE AND CHECK 05.P1** After present value of a single sum, state the rule in your own words.

**PAUSE AND CHECK 05.P2** After the mixed stream, state the rule in your own words.



*a clerk counting out regular payments at a post-office window with a queue, 1880s*

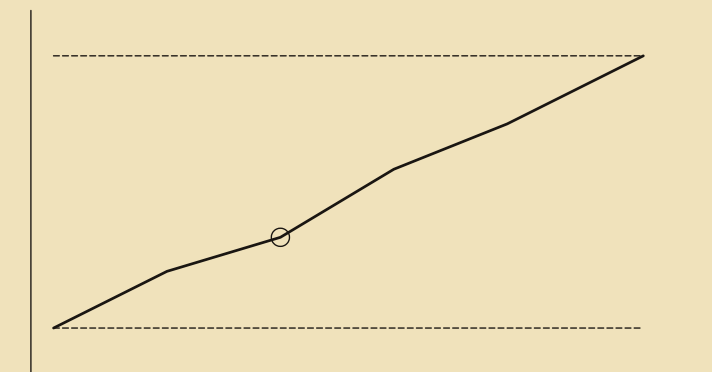
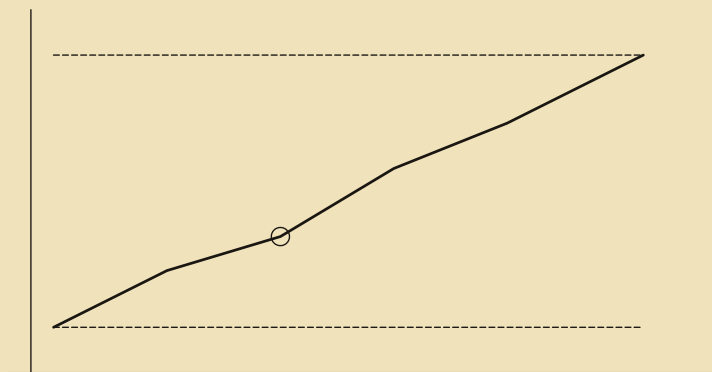
## CHAPTER VI

# Annuities

*A practical question about annuities*



*This chapter introduces annuities for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 6.1. *Annuities figure 1*FIG. 6.2. *Annuities figure 2*

*This section builds the idea of what an annuity is before moving to application.*

**Annuity.** A series of equal payments made at equal intervals for a fixed number of periods..

*The term annuity gives the chapter a precise name for this part of the problem.*

*The central question in what an annuity is is what changes when the underlying decision changes.*

## 6.1 WHAT AN ANNUITY IS

The purpose here is to make what an annuity is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Annuity** means A series of equal payments made at equal intervals for a fixed number of periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what an annuity is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 6.2 FUTURE VALUE OF AN ANNUITY

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The purpose here is to make future value of an annuity concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Ordinary annuity** means An annuity whose payments fall at the end of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For future value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## 6.3 PRESENT VALUE OF AN ANNUITY

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The purpose here is to make present value of an annuity concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of future value of an annuity before moving to application.*

**Ordinary annuity.** An annuity whose payments fall at the end of each period..

*The term ordinary annuity gives the chapter a precise name for this part of the problem.*

*The central question in future value of an annuity is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of present value of an annuity before moving to application.*

**Annuity due.** An annuity whose payments fall at the start of each period..

*The term annuity due gives the chapter a precise name for this part of the problem.*

*The central question in present value of an annuity is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

In this chapter, **Annuity due** means An annuity whose payments fall at the start of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For present value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 6.4 SOLVING FOR THE PAYMENT

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*This section builds the idea of solving for the payment before moving to application.*

**Instalment.** One of a series of regular payments used to repay a loan or build a sum..

*The term instalment gives the chapter a precise name for this part of the problem.*

*The central question in solving for the payment is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

The purpose here is to make solving for the payment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Instalment** means One of a series of regular payments used to repay a loan or build a sum. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For solving for the payment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to annuities. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

each method begins with a disciplined description of the cash or value being studied.

*The chapter has connected its main terms, method, and business decision.*

*Finance becomes stronger when a number is followed by a question about meaning.*

*Most errors arise from mixing terms, timing, signs, or decision rules.*

*A historical case shows how the chapter's financial problem appeared in practice.*

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## SUMMARY

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The central purpose of Annuities is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

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## THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

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## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

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## IN HISTORY

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Johan de Witt, 1671 report to the States of Holland on valuing life annuities for the Dutch Republic. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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## REVIEW QUESTIONS

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**06.T1** say what makes a series of payments an annuity and name its two timing forms

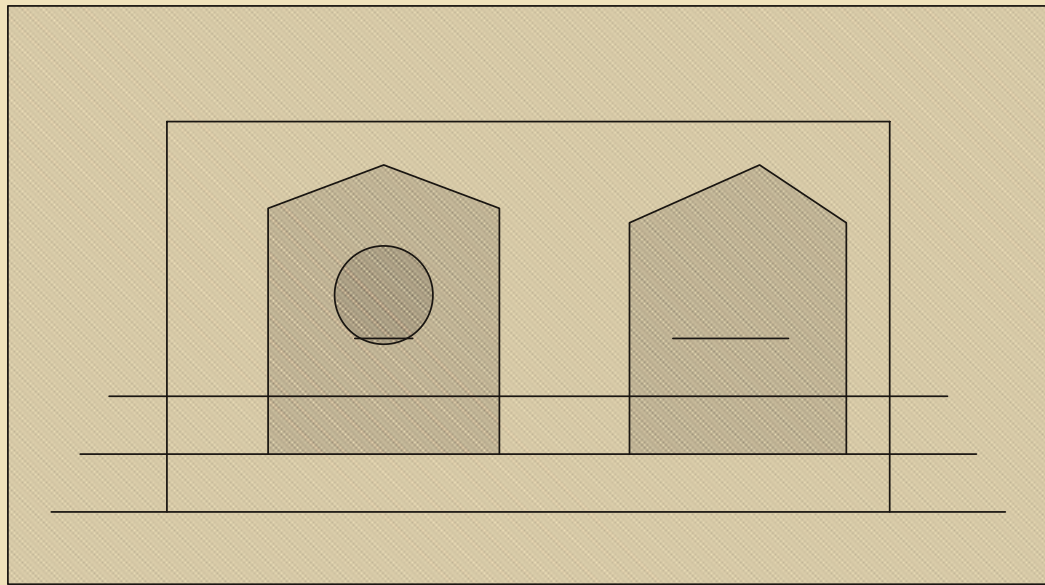
**06.T2** find the future value of an annuity for several rate and period pairs as ordinary and as due

**06.T3** find the present value of a new instalment plan

**06.T4** compute the yearly deposit needed for a stated goal

**PAUSE AND CHECK 06.P1** After annuity types, state the rule in your own words.

**PAUSE AND CHECK 06.P2** After future value of an ordinary annuity, state the rule in your own words.



*an engineer unrolling the plans of a railway bridge on a drawing table, 1860s*

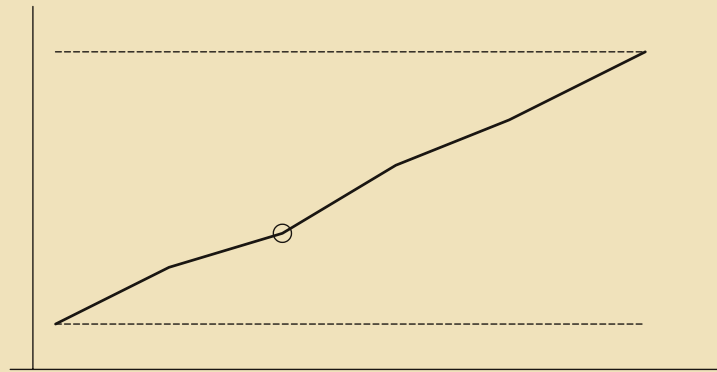
## CHAPTER VII

# Capital Budgeting: Process and Cash Flows

*A practical question about capital budgeting: process and cash flows*



*This chapter introduces capital budgeting: process and cash flows for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 7.1. *Capital Budgeting: Process and Cash Flows figure 1*

*This section builds the idea of why firms budget for capital before moving to application.*

**Capital budgeting.** Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring..

*The term capital budgeting gives the chapter a precise name for this part of the problem.*

**Capital expenditure.** Spending on an asset that will help a business for more than one year..

*The term capital expenditure gives the chapter a precise name for this part of the problem.*

**Independent project.** A project that can be accepted or rejected without affecting any other project..

*The term independent project gives the chapter a precise name for this part of the problem.*

**Mutually exclusive projects.** Projects of which only one can be chosen, because choosing one rules out the other..

*The term mutually exclusive projects gives the chapter a precise name for this part of the problem.*

*The central question in why firms budget for capital is what changes when the underlying decision changes.*

## 7.1 WHY FIRMS BUDGET FOR CAPITAL

The purpose here is to make why firms budget for capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital budgeting** means Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital expenditure** means Spending on an asset that will help a business for more than one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Independent project** means A project that can be accepted or rejected without affecting any other project. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Mutually exclusive projects** means Projects of which only one can be chosen, because choosing one rules out the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why firms budget for capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 7.2 THE STEPS OF THE PROCESS

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The purpose here is to make the steps of the process concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Relevant cash flow** means A cash flow that changes because of the decision being studied. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sunk cost** means Money already spent that cannot be recovered whatever you decide now. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Initial investment** means The cash flows at the start of a project to buy and prepare the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Operating cash flow** means The cash a project brings in each year after paying its cash costs and tax. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of the steps of the process before moving to application.*

**Relevant cash flow.** A cash flow that changes because of the decision being studied..

*The term relevant cash flow gives the chapter a precise name for this part of the problem.*

**Sunk cost.** Money already spent that cannot be recovered whatever you decide now..

*The term sunk cost gives the chapter a precise name for this part of the problem.*

**Initial investment.** The cash flows at the start of a project to buy and prepare the asset..

*The term initial investment gives the chapter a precise name for this part of the problem.*

**Operating cash flow.** The cash a project brings in each year after paying its cash costs and tax..

*The term operating cash flow gives the chapter a precise name for this part of the problem.*

*The central question in the steps of the process is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

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### 7.3 CASH FLOWS FROM AN INVESTMENT

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*This section builds the idea of cash flows from an investment before moving to application.*

**Terminal cash flow.** The cash received or paid when a project ends, such as the sale of scrap..

*The term terminal cash flow gives the chapter a precise name for this part of the problem.*

**Depreciation.** The share of an asset's cost charged against income each year as the asset wears out..

*The term depreciation gives the chapter a precise name for this part of the problem.*

*The central question in cash flows from an investment is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

The purpose here is to make cash flows from an investment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Terminal cash flow** means The cash received or paid when a project ends, such as the sale of scrap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

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Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For cash flows from an investment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

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*A complete worked example joins the chapter ideas in one decision.*

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to capital budgeting: process and cash flows. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep

the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Capital Budgeting: Process and Cash Flows is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

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## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

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## IN HISTORY

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The Concorde programme: UK and France agreed in 1962 to build it and continued after costs soared, a famous example of sunk-cost thinking; service ended in 2003. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

*A historical case shows how the chapter's financial problem appeared in practice.*

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## REVIEW QUESTIONS

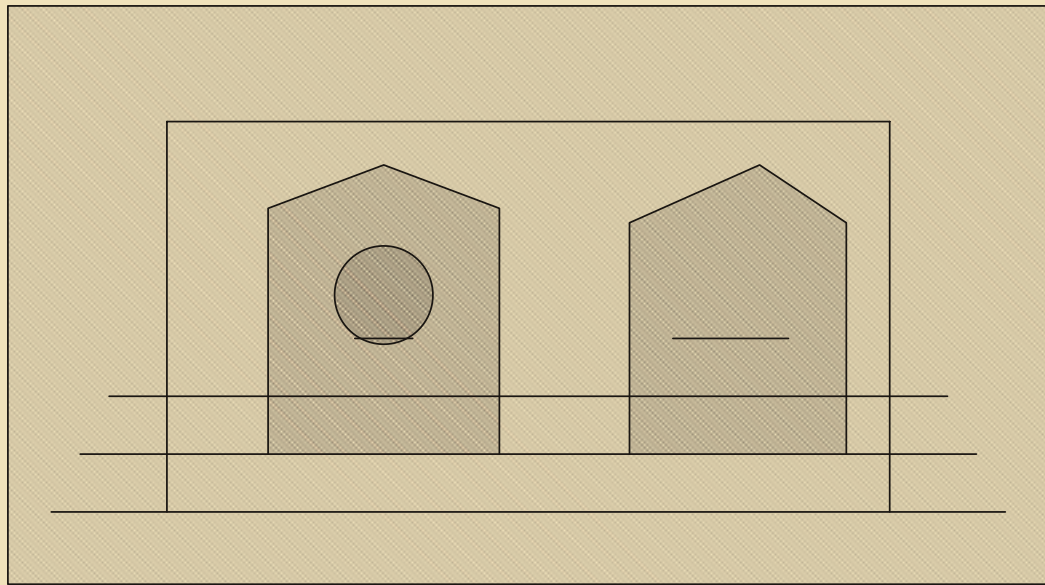
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- 07.T1 say why capital budgeting matters to a business, with a new example
- 07.T2 list the steps of the process in order and what each step produces
- 07.T3 sort a list of costs into relevant and sunk

**07.T4** find the operating cash flow from a new set of sales, costs, depreciation and tax

**PAUSE AND CHECK 07.P1** After the steps, state the rule in your own words.

**PAUSE AND CHECK 07.P2** After relevant cash flows, state the rule in your own words.



*a surveyor with a theodolite beside a canal under construction, 1850s*

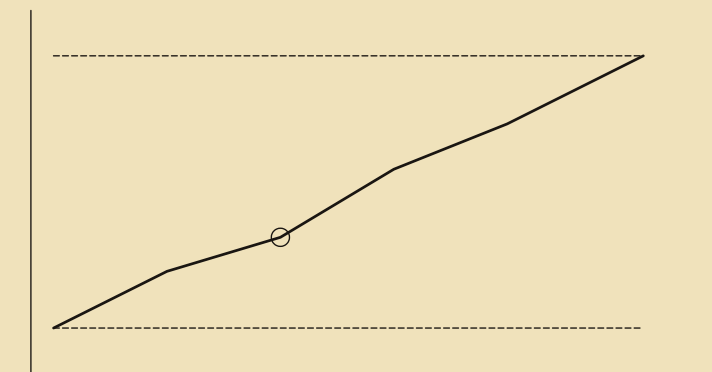
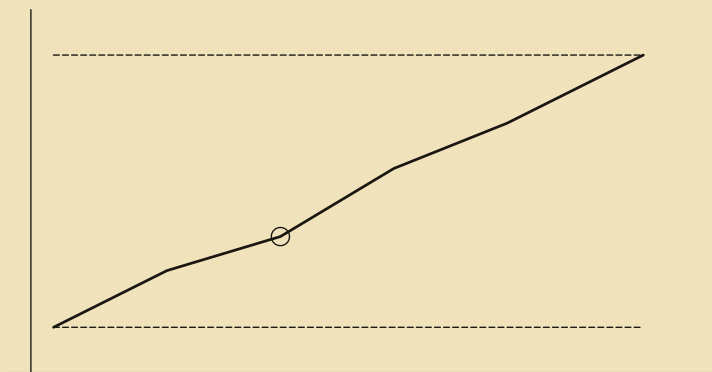
#### CHAPTER VIII

# Evaluating Projects: Payback and Net Present Value

*A practical question about evaluating projects: payback and net present value*



*This chapter introduces evaluating projects: payback and net present value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 8.1. *Evaluating Projects: Payback and Net Present Value figure 1*FIG. 8.2. *Evaluating Projects: Payback and Net Present Value figure 2*

*This section builds the idea of payback period before moving to application.*

**Payback period.** The time a project takes to return its initial investment from its cash flows..

*The term payback period gives the chapter a precise name for this part of the problem.*

**Discounted payback period.** The payback time when each cash flow is first discounted to present value..

*The term discounted payback period gives the chapter a precise name for this part of the problem.*

## 8.1 PAYBACK PERIOD

The purpose here is to make payback period concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Payback period** means The time a project takes to return its initial investment from its cash flows. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discounted payback period** means The payback time when each cash flow is first discounted to present value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For payback period, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## 8.2 NET PRESENT VALUE

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The purpose here is to make net present value concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Net present value** means The present value of all a project's future cash flows minus its initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Internal rate of return** means The discount rate at which a project's net present value is exactly zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For net present value, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

*The central question in payback period is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of net present value before moving to application.*

**Net present value.** The present value of all a project's future cash flows minus its initial investment..

*The term net present value gives the chapter a precise name for this part of the problem.*

**Internal rate of return.** The discount rate at which a project's net present value is exactly zero..

*The term internal rate of return gives the chapter a precise name for this part of the problem.*

*The central question in net present value is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

\* \* \*

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### 8.3 OTHER MEASURES OF A PROJECT

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*This section builds the idea of other measures of a project before moving to application.*

**Profitability index.** Present value of future cash flows divided by the initial investment..

*The term profitability index gives the chapter a precise name for this part of the problem.*

*The central question in other measures of a project is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

The purpose here is to make other measures of a project concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profitability index** means Present value of future cash flows divided by the initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For other measures of a project, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### 8.4 CHOOSING AMONG METHODS

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*This section builds the idea of choosing among methods before moving to application.*

*The central question in choosing among methods is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

The purpose here is to make choosing among methods concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For choosing among methods, keep the decision in view. Ask what the business is trying

to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

### WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to evaluating projects: payback and net present value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Evaluating Projects: Payback and Net Present Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

*Most errors arise from mixing terms, timing, signs, or decision rules.*

*A historical case shows how the chapter's financial problem appeared in practice.*

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### WHERE PEOPLE GO WRONG

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Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

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### IN HISTORY

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The Channel Tunnel, opened 6 May 1994, whose traffic forecasts and costs differed greatly from plan. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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### REVIEW QUESTIONS

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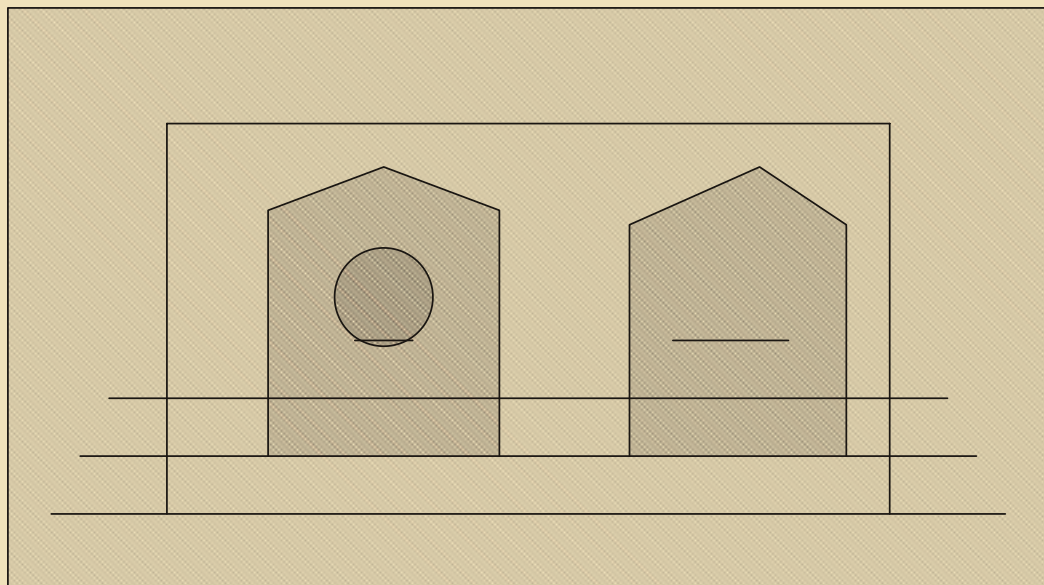
**08.T1** compute payback and net present value for new project data and state the decision rule for each

**08.T2** say what the internal rate of return means in plain words

**08.T3** rank three new projects by profitability index

**PAUSE AND CHECK 08.P1** After payback, state the rule in your own words.

**PAUSE AND CHECK 08.P2** After net present value, state the rule in your own words.



*a lighthouse on a rocky coast with a sailing ship in a rough sea, 1860s*

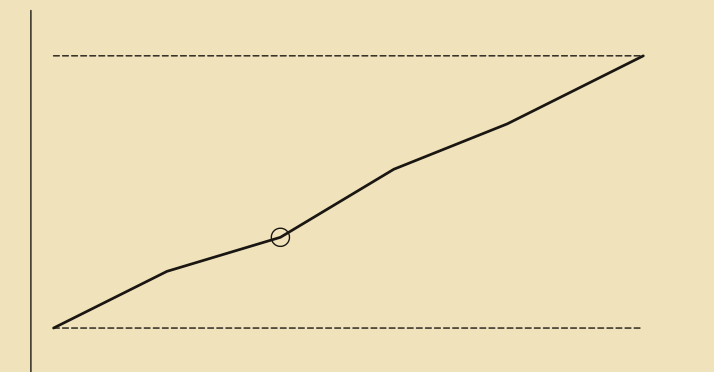
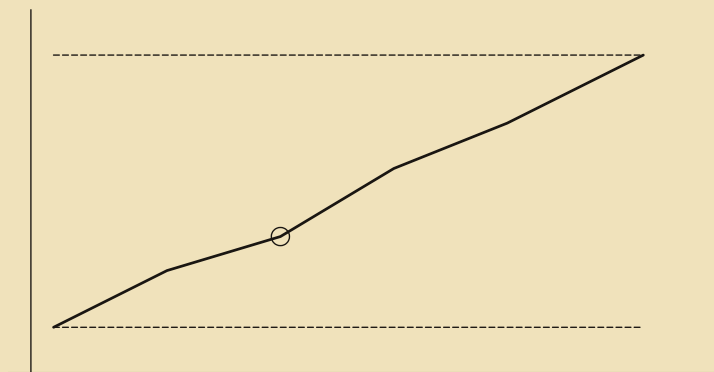
## CHAPTER IX

# Risk and Return

*A practical question about risk and return*



*This chapter introduces risk and return for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 9.1. *Risk and Return figure 1*FIG. 9.2. *Risk and Return figure 2*


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## 9.1 RETURN AND RISK

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*This section builds the idea of return and risk before moving to application.*

**Return.** The gain or loss from an investment, including income received and any change in value..

*The term return gives the chapter a precise name for this part of the problem.*

**Rate of return.** The return in one period divided by the sum invested, written as a percentage..

*The term rate of return gives the chapter a precise name for this part of the problem.*

The purpose here is to make return and risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Return** means The gain or loss from an investment, including income received and any change in value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Rate of return** means The return in one period divided by the sum invested, written as a percentage. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Risk** means The chance that the actual result will differ from the result you expect. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Business risk** means Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For return and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## 9.2 DESCRIBING UNCERTAINTY

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The purpose here is to make describing uncertainty concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial risk** means Extra risk that comes from borrowing, because interest must be paid whatever the sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Probability distribution** means A list of possible results with the chance of each, the chances adding to one. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

**Risk.** The chance that the actual result will differ from the result you expect..

*The term risk gives the chapter a precise name for this part of the problem.*

**Business risk.** Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing..

*The term business risk gives the chapter a precise name for this part of the problem.*

*The central question in return and risk is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of describing uncertainty before moving to application.*

**Financial risk.** Extra risk that comes from borrowing, because interest must be paid whatever the sales..

*The term financial risk gives the chapter a precise name for this part of the problem.*

**Probability distribution.** A list of possible results with the chance of each, the chances adding to one..

*The term probability distribution gives the chapter a precise name for this part of the problem.*

**Expected value.** The average of possible results, each weighted by its probability..

*The term expected value gives the chapter a precise name for this part of the problem.*

**Variance.** The probability-weighted average of the squared gaps between each result and the expected value..

*The term variance gives the chapter a precise name for this part of the problem.*

*The central question in describing uncertainty is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

In this chapter, **Expected value** means The average of possible results, each weighted by its probability. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Variance** means The probability-weighted average of the squared gaps between each result and the expected value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For describing uncertainty, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### 9.3 MEASURING RISK

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*This section builds the idea of measuring risk before moving to application.*

**Standard deviation.** The square root of the variance; it shows how far results typically stray from the expected value..

*The term standard deviation gives the chapter a precise name for this part of the problem.*

**Coefficient of variation.** Standard deviation divided by expected value, giving risk per unit of expected return..

*The term coefficient of variation gives the chapter a precise name for this part of the problem.*

*The central question in measuring risk is what changes when the underlying decision changes.*

The purpose here is to make measuring risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Standard deviation** means The square root of the variance; it shows how far results typically stray from the expected value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Coefficient of variation** means Standard deviation divided by expected value, giving risk per unit of expected return. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For measuring risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to risk and return. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*The chapter has connected its main terms, method, and business decision.*

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## SUMMARY

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The central purpose of Risk and Return is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

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## THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

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## WHERE PEOPLE GO WRONG

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Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

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## IN HISTORY

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Daniel Bernoulli, 1738 paper on measuring risk, which showed that people judge a gamble by its meaning to them as well as by its average. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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## REVIEW QUESTIONS

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**09.T1** define risk and say how business risk differs from financial risk, using a new firm

**09.T2** compute the expected value, then the variance, the standard deviation and the coefficient of variation, for a new two-asset table and state which is riskier per unit of return

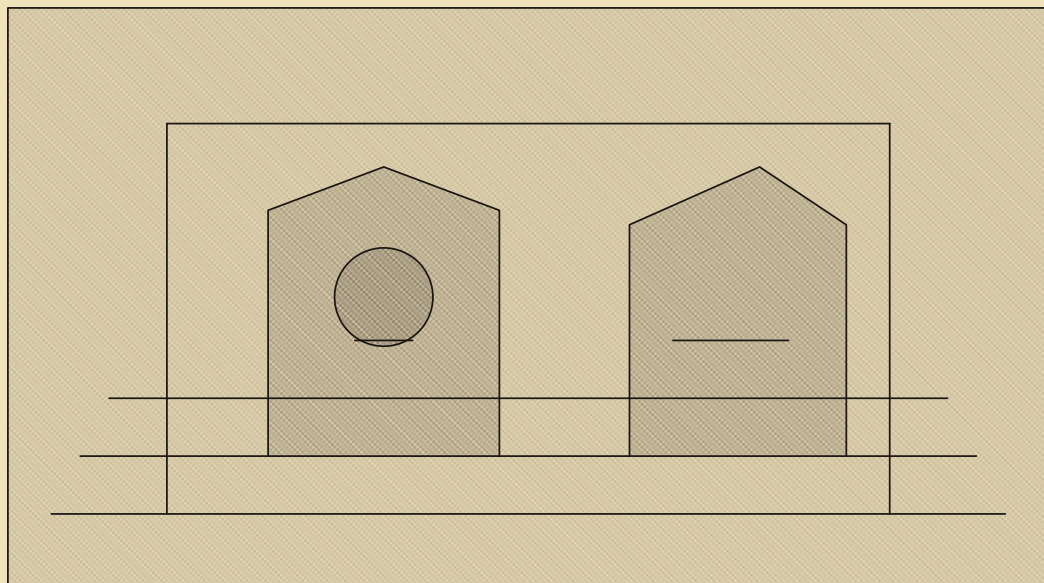
**09.T3** explain standard deviation in plain words

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**PAUSE AND CHECK 09.P1** After probability distribution, state the rule in your own words.

**PAUSE AND CHECK 09.P2** After standard deviation, state the rule in your own words.





*a long iron lever lifting a heavy stone block in a quarry, 1870s*

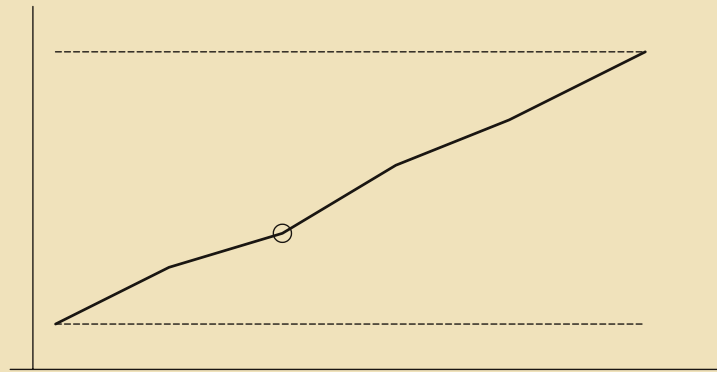
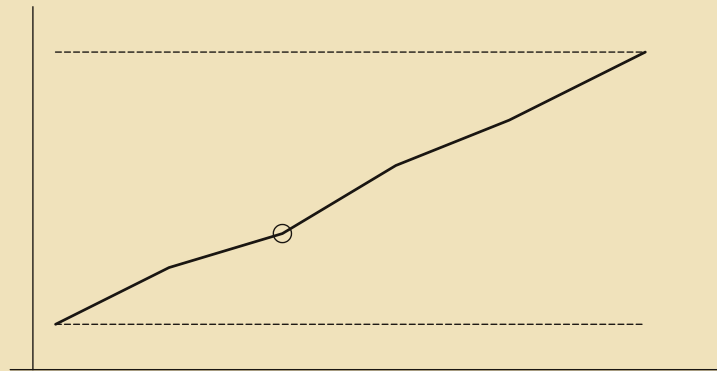
## CHAPTER X

# Leverage

*A practical question about leverage*



*This chapter introduces leverage for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 10.1. *Leverage figure 1*FIG. 10.2. *Leverage figure 2*

*This section builds the idea of costs and the break-even point before moving to application.*

**Leverage.** The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings..

*The term leverage gives the chapter a precise name for this part of the problem.*

**Fixed cost.** A cost that stays the same however many units are sold, such as shop rent..

*The term fixed cost gives the chapter a precise name for this part of the problem.*

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## 10.1 COSTS AND THE BREAK-EVEN POINT

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The purpose here is to make costs and the break-even point concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Leverage** means The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Fixed cost** means A cost that stays the same however many units are sold, such as shop rent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Variable cost** means A cost that rises and falls with the number of units sold, such as the price paid for each book. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For costs and the break-even point, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 10.2 OPERATING LEVERAGE

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The purpose here is to make operating leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Contribution margin** means Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Break-even point** means The sales volume at which total revenue equals total cost, so profit is zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **EBIT** means Earnings before interest and tax: sales minus operating costs. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

**Variable cost.** A cost that rises and falls with the number of units sold, such as the price paid for each book..

*The term variable cost gives the chapter a precise name for this part of the problem.*

*The central question in costs and the break-even point is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of operating leverage before moving to application.*

**Contribution margin.** Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit..

*The term contribution margin gives the chapter a precise name for this part of the problem.*

**Break-even point.** The sales volume at which total revenue equals total cost, so profit is zero..

*The term break-even point gives the chapter a precise name for this part of the problem.*

**EBIT.** Earnings before interest and tax: sales minus operating costs..

*The term ebit gives the chapter a precise name for this part of the problem.*

*The central question in operating leverage is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of financial leverage before moving to application.*

**Operating leverage.** The effect of fixed operating costs on how EBIT responds to sales.

*The term operating leverage gives the chapter a precise name for this part of the problem.*

**Financial leverage.** The effect of fixed financing payments on how owners' earnings respond to EBIT.

*The term financial leverage gives the chapter a precise name for this part of the problem.*

**Total leverage.** The combined effect of operating and financial leverage on how owners' earnings respond to sales.

*The term total leverage gives the chapter a precise name for this part of the problem.*

*The central question in financial leverage is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For operating leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### 10.3 FINANCIAL LEVERAGE

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The purpose here is to make financial leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating leverage** means The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the percentage change in EBIT divided by the percentage change in sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial leverage** means The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Total leverage** means The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For financial leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a

larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

## 10.4 TOTAL LEVERAGE AND RISK

The purpose here is to make total leverage and risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Earnings per share** means Earnings left for owners of common stock divided by the number of shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Preferred stock** means Shares that pay a fixed dividend, which must be paid before any dividend on common stock. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For total leverage and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

### WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to leverage. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000.

*This section builds the idea of total leverage and risk before moving to application.*

**Earnings per share.** Earnings left for owners of common stock divided by the number of shares..

*The term earnings per share gives the chapter a precise name for this part of the problem.*

**Preferred stock.** Shares that pay a fixed dividend, which must be paid before any dividend on common stock..

*The term preferred stock gives the chapter a precise name for this part of the problem.*

*The central question in total leverage and risk is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Leverage is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

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## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

*A historical case shows how the chapter's financial problem appeared in practice.*

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## IN HISTORY

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Long-Term Capital Management, whose heavy borrowing led to a rescue organised by the Federal Reserve Bank of New York on 23 September 1998. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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## REVIEW QUESTIONS

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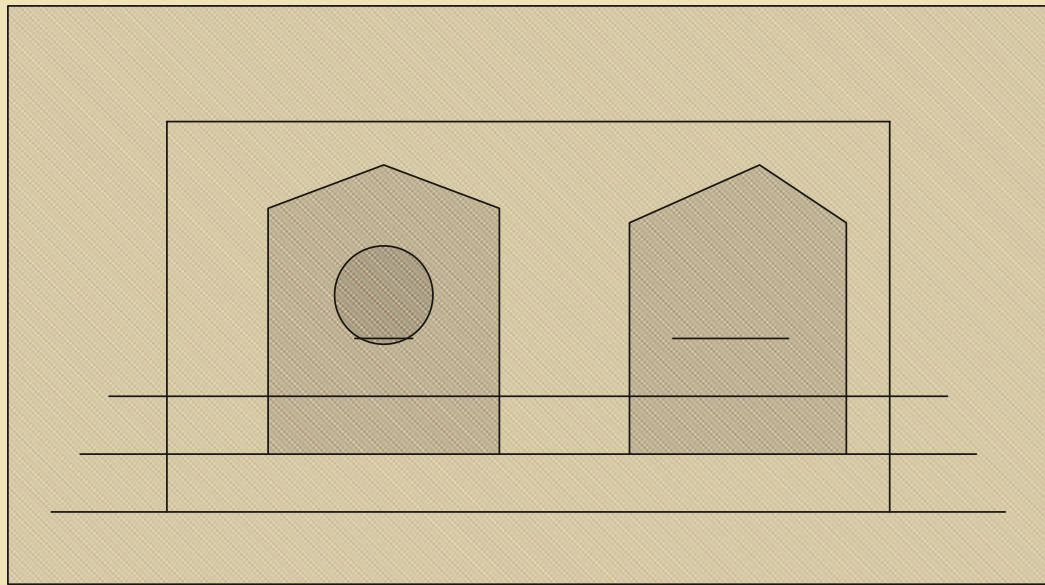
**10.T1** say what leverage means and tell its three kinds apart with new data

**10.T2** find the break-even point and degrees of operating, financial and total leverage for a new firm

**10.T3** explain in plain words why high leverage raises risk

**PAUSE AND CHECK 10.P1** After break-even point, state the rule in your own words.

**PAUSE AND CHECK 10.P2** After degree of operating leverage, state the rule in your own words.



*a cartographer's table with a coast chart, dividers and a brass compass, 1870s*

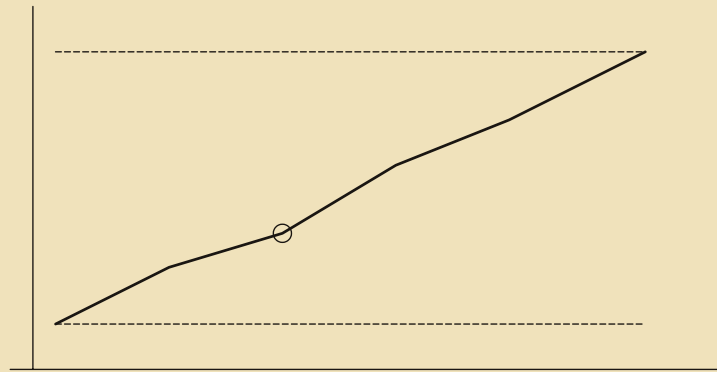
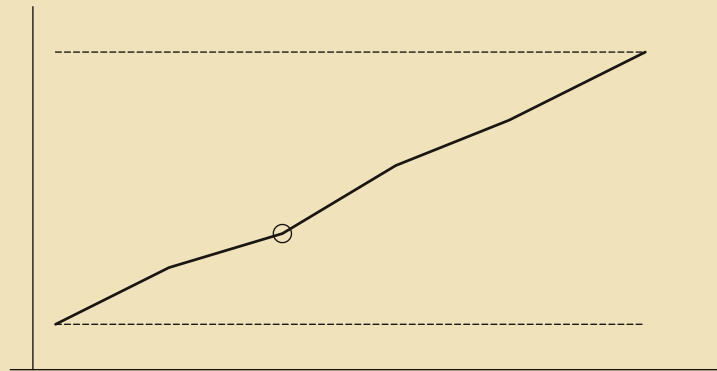
## CHAPTER XI

# Required Return, CAPM and Market Efficiency

*A practical question about required return, capm and market efficiency*



*This chapter introduces required return, capm and market efficiency for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 11.1. *Required Return, CAPM and Market Efficiency figure 1*FIG. 11.2. *Required Return, CAPM and Market Efficiency figure 2*

*This section builds the idea of risk premium and required return before moving to application.*

**Required return.** The least return an investor will accept for taking a given risk..

*The term required return gives the chapter a precise name for this part of the problem.*

**Risk-free rate.** The return on an investment with no risk of loss, such as a short-term government security..

*The term risk-free rate gives the chapter a precise name for this part of the problem.*

## 11.1 RISK PREMIUM AND REQUIRED RETURN

The purpose here is to make risk premium and required return concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Required return** means The least return an investor will accept for taking a given risk. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Risk-free rate** means The return on an investment with no risk of loss, such as a short-term government security. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Risk premium** means The extra return demanded for taking risk instead of a risk-free investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For risk premium and required return, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 11.2 DIVERSIFICATION AND BETA

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The purpose here is to make diversification and beta concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Portfolio** means A collection of investments held together by one investor. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Diversification** means Spreading money across many investments so that one failure does less damage. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Systematic risk** means Risk that affects all investments, such as a recession; it cannot be diversified away. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

**Risk premium.** The extra return demanded for taking risk instead of a risk-free investment..

*The term risk premium gives the chapter a precise name for this part of the problem.*

*The central question in risk premium and required return is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of diversification and beta before moving to application.*

**Portfolio.** A collection of investments held together by one investor..

*The term portfolio gives the chapter a precise name for this part of the problem.*

**Diversification.** Spreading money across many investments so that one failure does less damage..

*The term diversification gives the chapter a precise name for this part of the problem.*

**Systematic risk.** Risk that affects all investments, such as a recession; it cannot be diversified away..

*The term systematic risk gives the chapter a precise name for this part of the problem.*

*The central question in diversification and beta is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of the capital asset pricing model before moving to application.*

**Unsystematic risk.** Risk that affects one firm or industry only; diversification can remove it..

*The term unsystematic risk gives the chapter a precise name for this part of the problem.*

**Beta.** A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market..

*The term beta gives the chapter a precise name for this part of the problem.*

**Market return.** The average return of all investments in a market during a period..

*The term market return gives the chapter a precise name for this part of the problem.*

*The central question in the capital asset pricing model is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For diversification and beta, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

### 11.3 THE CAPITAL ASSET PRICING MODEL

The purpose here is to make the capital asset pricing model concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Unsystematic risk** means Risk that affects one firm or industry only; diversification can remove it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Beta** means A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Market return** means The average return of all investments in a market during a period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the capital asset pricing model, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a

later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 11.4 EFFICIENT MARKETS

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The purpose here is to make efficient markets concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital asset pricing model** means A formula for required return: risk-free rate plus beta times the market risk premium. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Security market line** means The graph of the capital asset pricing model, with beta on one axis and required return on the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Efficient market** means A market in which prices already reflect the available information. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For efficient markets, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to required return, capm and market efficiency. Start by listing the known amounts and their dates. Separate cash inflows from

*This section builds the idea of efficient markets before moving to application.*

**Capital asset pricing model.** A formula for required return: risk-free rate plus beta times the market risk premium..

*The term capital asset pricing model gives the chapter a precise name for this part of the problem.*

**Security market line.** The graph of the capital asset pricing model, with beta on one axis and required return on the other..

*The term security market line gives the chapter a precise name for this part of the problem.*

**Efficient market.** A market in which prices already reflect the available information..

*The term efficient market gives the chapter a precise name for this part of the problem.*

*The central question in efficient markets is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Required Return, CAPM and Market Efficiency is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

---

## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

*Most errors arise from mixing terms, timing, signs, or decision rules.*

before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*A historical case shows how the chapter's financial problem appeared in practice.*

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### IN HISTORY

---

William Sharpe's 1964 paper on capital asset prices, and the 1990 Nobel Prize in economics shared by Markowitz, Miller and Sharpe. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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### SOURCES

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*The chapter points back to related ideas elsewhere in the book.*

Chapter 9, 'Risk and Return', section 9.3 'Measuring risk' — standard deviation as the risk measure; Chapter 13, 'Valuation of Shares', section 13.3 'Valuing common stock' — required return used as the discount rate; Chapter 15, 'Long-Term Financing and the Capital Market', section 15.4 'The cost of capital' — equity cost..

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### REVIEW QUESTIONS

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**11.T1** explain risk premium and why investors demand it, with new figures

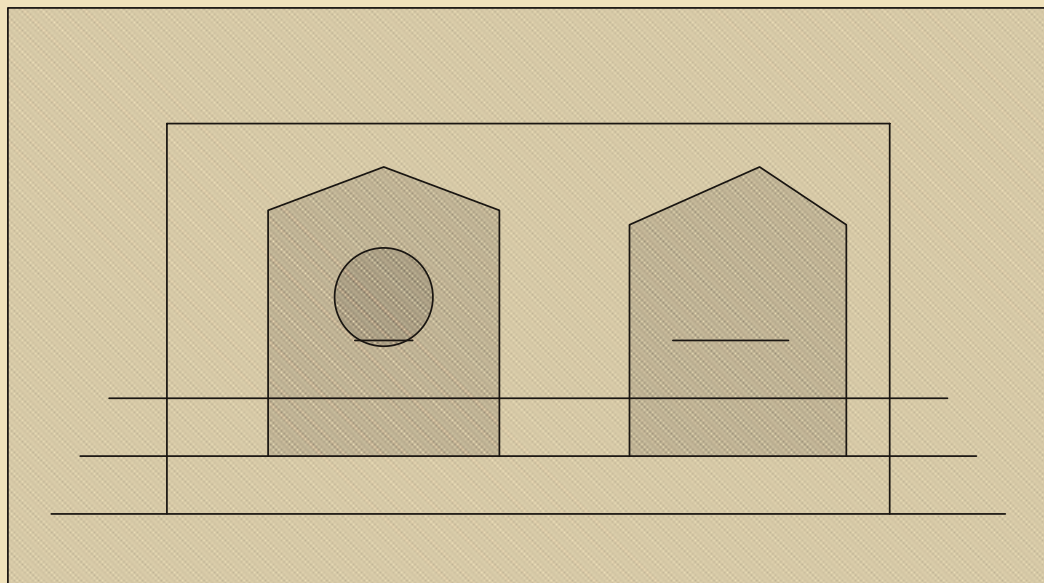
**11.T2** separate systematic from unsystematic risk using a new list of risks

**11.T3** compute required return for three new betas

**11.T4** describe the three forms of market efficiency and what each implies

**PAUSE AND CHECK 11.P1** After risk premium, state the rule in your own words.

**PAUSE AND CHECK 11.P2** After beta, state the rule in your own words.



*a bundle of bonds tied with ribbon and a wax seal on a lawyer's desk, 1880s*

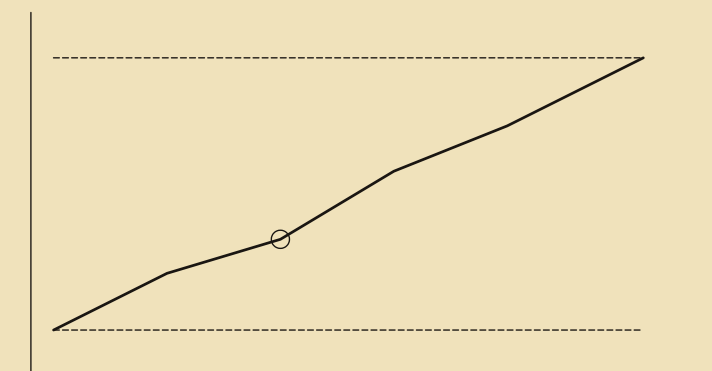
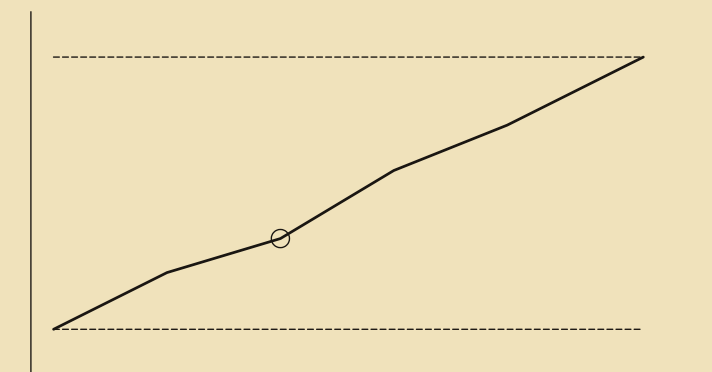
## CHAPTER XII

# Valuation of Bonds

*A practical question about valuation of bonds*



*This chapter introduces valuation of bonds for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 12.1. *Valuation of Bonds figure 1*FIG. 12.2. *Valuation of Bonds figure 2*

*This section builds the idea of the idea of valuation before moving to application.*

**Valuation.** Finding what an asset is worth today by adding the present values of the cash it will bring..

*The term valuation gives the chapter a precise name for this part of the problem.*

---

## 12.1 THE IDEA OF VALUATION

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The purpose here is to make the idea of valuation concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Valuation** means Finding what an asset is worth today by adding the present values of the cash it will bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Bond** means A loan made to a company or government, repaid on a set date, with interest paid on the way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Face value** means The amount a bond repays at its end, for example Tk 1,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the idea of valuation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 12.2 WHAT A BOND IS

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The purpose here is to make what a bond is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Coupon rate** means The yearly interest of a bond, as a percentage of its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Maturity** means The date on which a bond or loan must be repaid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Yield to maturity** means The yearly return an investor earns by buying a bond at today's price and holding it to maturity. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

**Bond.** A loan made to a company or government, repaid on a set date, with interest paid on the way..

*The term bond gives the chapter a precise name for this part of the problem.*

**Face value.** The amount a bond repays at its end, for example Tk 1,000..

*The term face value gives the chapter a precise name for this part of the problem.*

*The central question in the idea of valuation is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of what a bond is before moving to application.*

**Coupon rate.** The yearly interest of a bond, as a percentage of its face value..

*The term coupon rate gives the chapter a precise name for this part of the problem.*

**Maturity.** The date on which a bond or loan must be repaid..

*The term maturity gives the chapter a precise name for this part of the problem.*

**Yield to maturity.** The yearly return an investor earns by buying a bond at today's price and holding it to maturity..

*The term yield to maturity gives the chapter a precise name for this part of the problem.*

*The central question in what a bond is is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what a bond is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### 12.3 PRICING A BOND

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*This section builds the idea of pricing a bond before moving to application.*

**Premium bond.** A bond that sells for more than its face value..

*The term premium bond gives the chapter a precise name for this part of the problem.*

**Discount bond.** A bond that sells for less than its face value..

*The term discount bond gives the chapter a precise name for this part of the problem.*

**Semiannual interest.** Interest paid twice a year, each payment being half the yearly amount..

*The term semiannual interest gives the chapter a precise name for this part of the problem.*

*The central question in pricing a bond is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

The purpose here is to make pricing a bond concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Premium bond** means A bond that sells for more than its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discount bond** means A bond that sells for less than its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Semiannual interest** means Interest paid twice a year, each payment being half the yearly amount. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For pricing a bond, keep the decision in view. Ask what the business is trying to buy,

finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

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## 12.4 YIELD AND PRICE

---

The purpose here is to make yield and price concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For yield and price, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## WORKED EXAMPLE

---

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of bonds. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

*This section builds the idea of yield and price before moving to application.*

*The central question in yield and price is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*



A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

*A short restatement keeps the financial meaning attached to the calculation.*

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Valuation of Bonds is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

*Finance becomes stronger when a number is followed by a question about meaning.*

*Most errors arise from mixing terms, timing, signs, or decision rules.*

*A historical case shows how the chapter's financial problem appeared in practice.*

---

### THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

---

### WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

---

### IN HISTORY

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British consols, perpetual bonds first issued in the 1750s, and the UK government's redemption of its 3.5% War Loan in 2015. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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### REVIEW QUESTIONS

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**12.T1** explain what a bond is and name its main features with a new issue

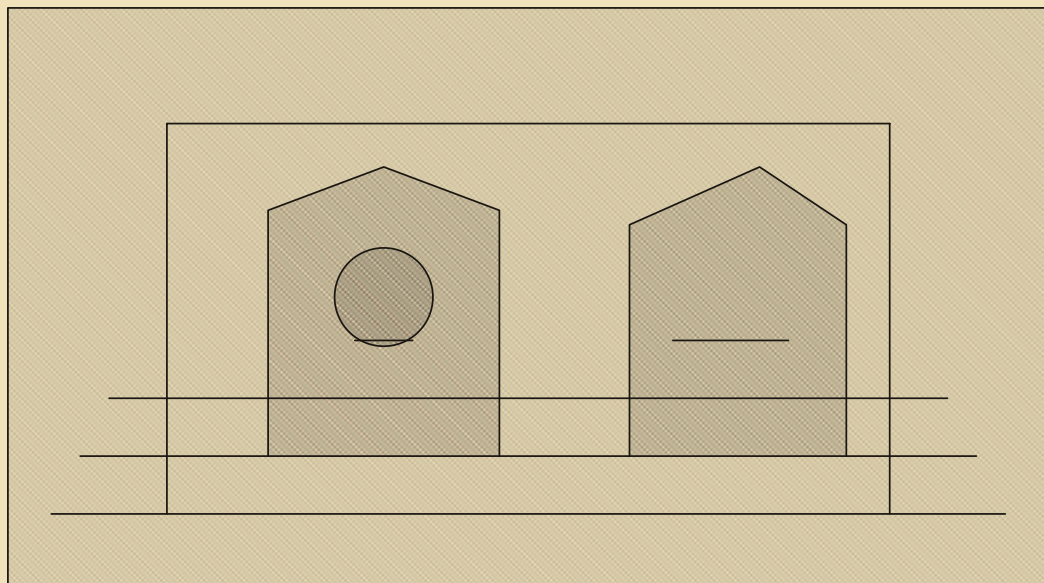
**12.T2** compute the value of three new bonds paying interest yearly and say whether each is at a premium, discount or par

**12.T3** explain the link between coupon rate, required return and price

**12.T4** define yield to maturity in plain words

**PAUSE AND CHECK 12.P1** After bond features, state the rule in your own words.

**PAUSE AND CHECK 12.P2** After pricing, state the rule in your own words.



*a stock exchange floor with brokers and a chalk board of prices, 1890s*

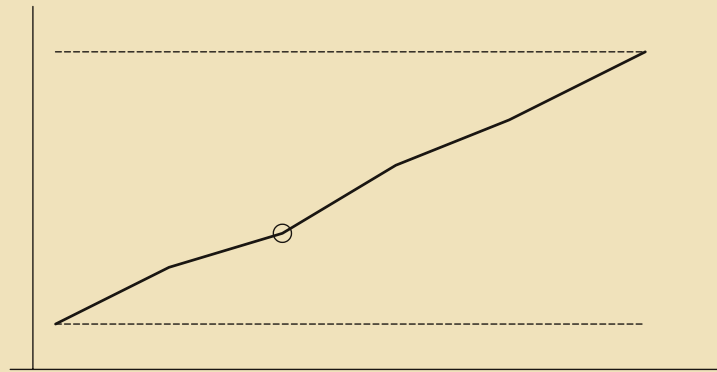
## CHAPTER XIII

# Valuation of Shares

*A practical question about valuation of shares*



*This chapter introduces valuation of shares for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 13.1. *Valuation of Shares figure 1*

*This section builds the idea of common stock before moving to application.*

**Common stock.** Shares that give owners a vote and a claim on profit after all other claims are met..

*The term common stock gives the chapter a precise name for this part of the problem.*

**Residual claim.** The right to whatever is left after lenders and preferred owners have been paid..

*The term residual claim gives the chapter a precise name for this part of the problem.*

*The central question in common stock is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

### 13.1 COMMON STOCK

The purpose here is to make common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Common stock** means Shares that give owners a vote and a claim on profit after all other claims are met. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Residual claim** means The right to whatever is left after lenders and preferred owners have been paid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For common stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 13.2 VALUING PREFERRED STOCK

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The purpose here is to make valuing preferred stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital gain** means The profit from selling an asset for more than its purchase price. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Perpetuity** means A series of equal payments that continues for ever. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing preferred stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 13.3 VALUING COMMON STOCK

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The purpose here is to make valuing common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend valuation model** means A way of valuing a share as the present value of the dividends it is expected to pay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

*This section builds the idea of valuing preferred stock before moving to application.*

**Capital gain.** The profit from selling an asset for more than its purchase price..

*The term capital gain gives the chapter a precise name for this part of the problem.*

**Perpetuity.** A series of equal payments that continues for ever..

*The term perpetuity gives the chapter a precise name for this part of the problem.*

*The central question in valuing preferred stock is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of valuing common stock before moving to application.*

**Dividend valuation model.** A way of valuing a share as the present value of the dividends it is expected to pay..

*The term dividend valuation model gives the chapter a precise name for this part of the problem.*

**Constant growth model.** The dividend valuation model for dividends that grow at the same rate every year..

*The term constant growth model gives the chapter a precise name for this part of the problem.*

*The central question in valuing common stock is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing common stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of shares. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A short restatement keeps the financial meaning attached to the calculation.*

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### SUMMARY

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The central purpose of Valuation of Shares is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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### THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

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### WHERE PEOPLE GO WRONG

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Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

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### IN HISTORY

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The South Sea Company of 1720: its share price rose and collapsed in one year while its earnings stayed small. Valuation theory: John Burr Williams, *The Theory of Investment Value* (1938) and Myron Gordon (1959). The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

*A historical case shows how the chapter's financial problem appeared in practice.*

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### SOURCES

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Chapter 12, 'Valuation of Bonds', section 12.1 'The idea of valuation' — the principle used; Chapter 11, 'Required Return, CAPM and Market Efficiency', section 11.3 'The capital asset pricing model' — the required return; Chapter 15, 'Long-Term Financing and the Capital Market', section 15.2 'Debt and equity as sources' — stock as a source of money..

*The chapter points back to related ideas elsewhere in the book.*

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### REVIEW QUESTIONS

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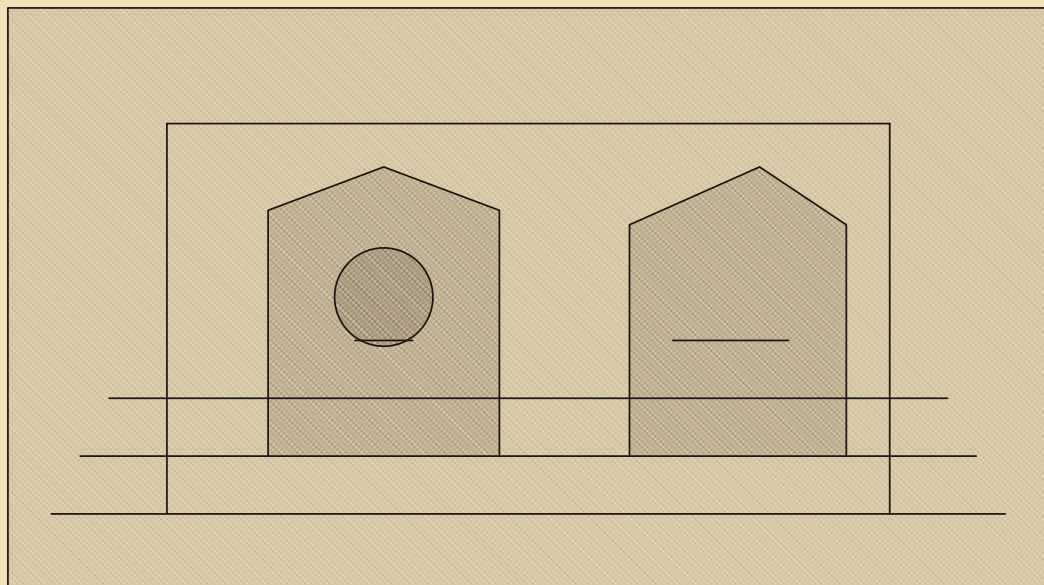
13.T1 say how common stock differs from a bond using a new firm

**13.T2** value a preferred stock and a growing common stock from new data

**13.T3** explain why growth raises value but also needs a required return above growth

**PAUSE AND CHECK 13.P1** After common stock, state the rule in your own words.

**PAUSE AND CHECK 13.P2** After perpetuity, state the rule in your own words.



*a warehouse with sacks, crates and a weighing machine watched by a clerk, 1880s*

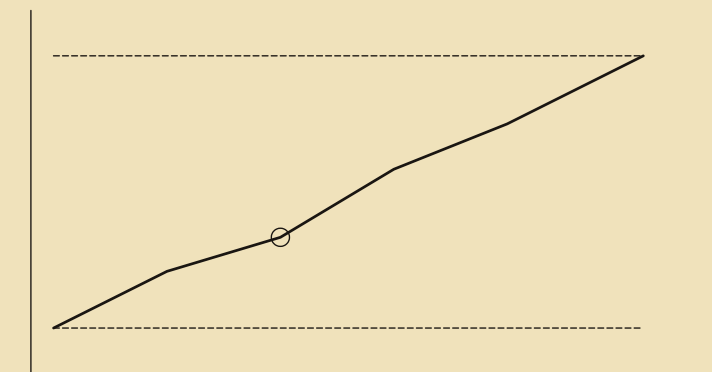
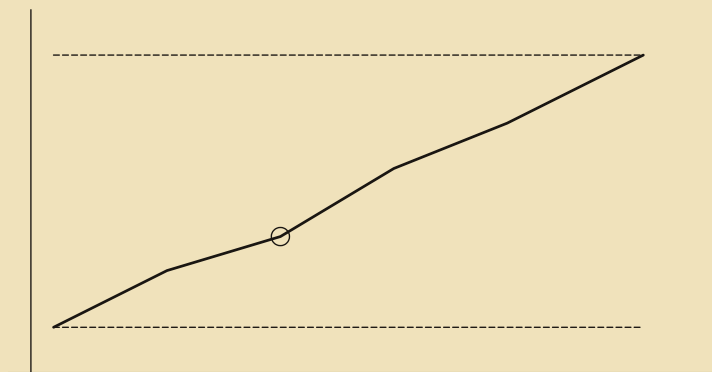
CHAPTER XIV

# Working Capital and Short-Term Financing

*A practical question about working capital and short-term financing*



*This chapter introduces working capital and short-term financing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 14.1. *Working Capital and Short-Term Financing figure 1*FIG. 14.2. *Working Capital and Short-Term Financing figure 2*


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## 14.1 WORKING CAPITAL AND HOW TO FINANCE IT

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*This section builds the idea of working capital and how to finance it before moving to application.*

**Working capital.** The money a business has tied up in everyday trading: cash, stock and money owed to it..

*The term working capital gives the chapter a precise name for this part of the problem.*

The purpose here is to make working capital and how to finance it concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Working capital** means The money a business has tied up in everyday trading: cash, stock and money owed to it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Net working capital** means Current assets minus current liabilities; what remains for trading after short debts are paid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Matching principle** means Pay for short-lived needs with short-term money and for lasting needs with long-term money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Short-term financing** means Money borrowed to be repaid within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Intermediate-term financing** means Money borrowed to be repaid in more than one year but not more than about five. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For working capital and how to finance it, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

**Net working capital.** Current assets minus current liabilities; what remains for trading after short debts are paid..

*The term net working capital gives the chapter a precise name for this part of the problem.*

**Matching principle.** Pay for short-lived needs with short-term money and for lasting needs with long-term money..

*The term matching principle gives the chapter a precise name for this part of the problem.*

**Short-term financing.** Money borrowed to be repaid within one year..

*The term short-term financing gives the chapter a precise name for this part of the problem.*

**Intermediate-term financing.** Money borrowed to be repaid in more than one year but not more than about five..

*The term intermediate-term financing gives the chapter a precise name for this part of the problem.*

*The central question in working capital and how to finance it is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

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## 14.2 SOURCES OF SHORT-TERM FINANCE

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The purpose here is to make sources of short-term finance concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

*This section builds the idea of sources of short-term finance before moving to application.*

**Long-term financing.** Money raised for more than about five years, or with no fixed repayment date..

*The term long-term financing gives the chapter a precise name for this part of the problem.*

**Trade credit.** Credit from a supplier who lets a buyer pay some days after delivery..

*The term trade credit gives the chapter a precise name for this part of the problem.*

**Cash discount.** A price cut a supplier gives for paying early, such as 2% off if paid within 10 days..

*The term cash discount gives the chapter a precise name for this part of the problem.*

**Inventory.** The goods a business holds for sale, such as the books on a shop's shelves..

*The term inventory gives the chapter a precise name for this part of the problem.*

**Accounts receivable.** Money that customers owe a business for goods already delivered..

*The term accounts receivable gives the chapter a precise name for this part of the problem.*

*The central question in sources of short-term finance is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

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Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For sources of short-term finance, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### 14.3 CYCLES, FLOAT AND ESTIMATING NEEDS

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*This section builds the idea of cycles, float and estimating needs before moving to application.*

The purpose here is to make cycles, float and estimating needs concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the

chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Accounts payable** means Money a business owes its suppliers for goods already received. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Operating cycle** means The days from buying stock to collecting cash from its sale. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash conversion cycle** means The operating cycle minus the days a business takes to pay its suppliers. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Float** means Money tied up in payments that are sent but not yet usable, such as a cheque in the post. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For cycles, float and estimating needs, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to working capital and short-term financing. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

**Accounts payable.** Money a business owes its suppliers for goods already received..

*The term accounts payable gives the chapter a precise name for this part of the problem.*

**Operating cycle.** The days from buying stock to collecting cash from its sale..

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**Cash conversion cycle.** The operating cycle minus the days a business takes to pay its suppliers..

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**Float.** Money tied up in payments that are sent but not yet usable, such as a cheque in the post..

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*The central question in cycles, float and estimating needs is what changes when the underlying decision changes.*

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*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

*A short restatement keeps the financial meaning attached to the calculation.*

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Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$   $(1 + \text{rate})$ . Steps:  $\text{Tk } 100,000 \times (1 + 0.10) = \text{Tk } 110,000$ . Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*The chapter has connected its main terms, method, and business decision.*

*Finance becomes stronger when a number is followed by a question about meaning.*

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## SUMMARY

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The central purpose of Working Capital and Short-Term Financing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

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## THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

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## WHERE PEOPLE GO WRONG

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Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

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## IN HISTORY

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Toyota's just-in-time system, developed from the late 1940s by Taiichi Ohno and others to cut stock held. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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## REVIEW QUESTIONS

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**14.T1** say what working capital is and why a profitable firm can still run short

**14.T2** compute the cost of giving up a cash discount for new terms and recommend the cheaper source

**14.T3** find the cash conversion cycle and the financing need for new data

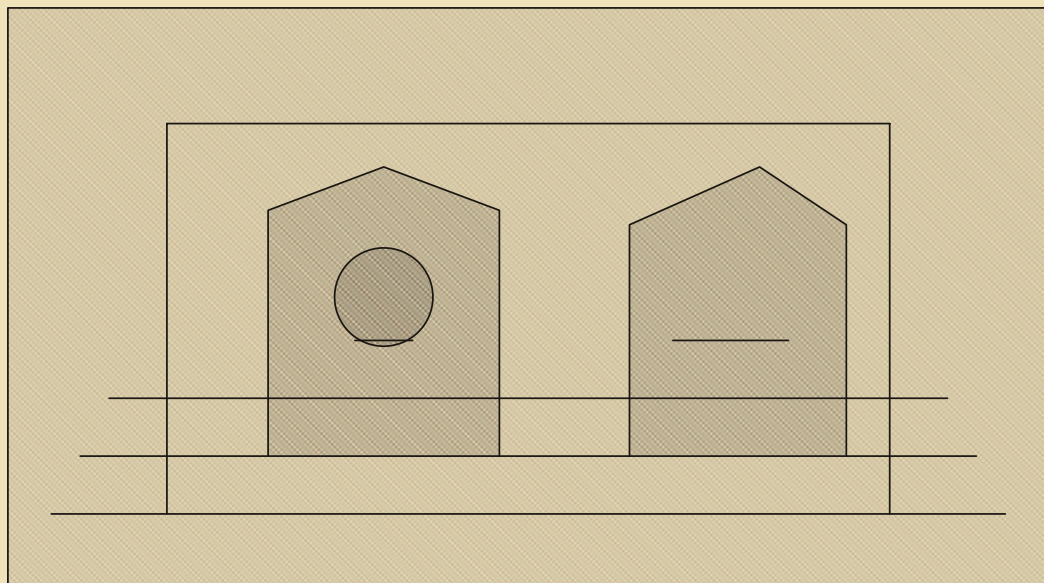
**14.T4** define float and its parts in plain words

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**PAUSE AND CHECK 14.P1** After matching principle, state the rule in your own words.

**PAUSE AND CHECK 14.P2** After cash conversion cycle, state the rule in your own words.





*an iron foundry with a glowing furnace and men pouring molten metal, 1870s*

CHAPTER XV

# Long-Term Financing and the Capital Market

*A practical question about long-term financing and the capital market*



*This chapter introduces long-term financing and the capital market for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

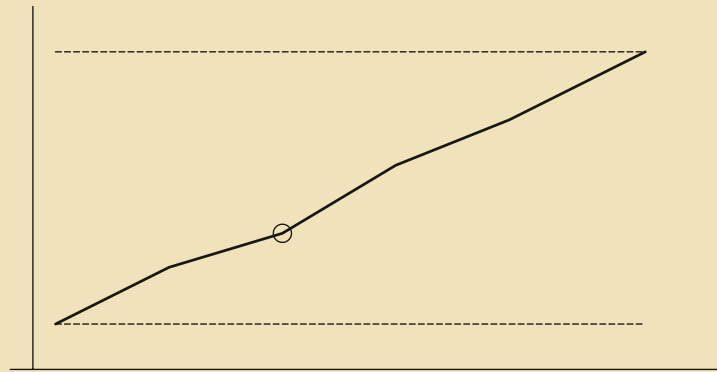


FIG. 15.1. Long-Term Financing and the Capital Market figure 1

### 15.1 TERM LOANS AND THE FEATURES OF FINANCING

*This section builds the idea of term loans and the features of financing before moving to application.*

**Term loan.** A loan repaid in regular instalments over a fixed number of years, usually from a bank..

*The term term loan gives the chapter a precise name for this part of the problem.*

**Collateral.** An asset a lender may take if the borrower fails to repay..

*The term collateral gives the chapter a precise name for this part of the problem.*

**Covenant.** A condition in a loan agreement that the borrower promises to keep..

*The term covenant gives the chapter a precise name for this part of the problem.*

*The central question in term loans and the features of financing is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

The purpose here is to make term loans and the features of financing concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Term loan** means A loan repaid in regular instalments over a fixed number of years, usually from a bank. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Collateral** means An asset a lender may take if the borrower fails to repay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Covenant** means A condition in a loan agreement that the borrower promises to keep. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For term loans and the features of financing, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 15.2 DEBT AND EQUITY AS SOURCES

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The purpose here is to make debt and equity as sources concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Debenture** means A bond backed only by the issuer's general promise to pay, not by a particular asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Retained earnings** means Profit kept in the business instead of being paid out as dividends. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital market** means The market in which long-term money is raised by selling bonds and shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For debt and equity as sources, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

\* \* \*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of debt and equity as sources before moving to application.*

**Debenture.** A bond backed only by the issuer's general promise to pay, not by a particular asset..

*The term debenture gives the chapter a precise name for this part of the problem.*

**Retained earnings.** Profit kept in the business instead of being paid out as dividends..

*The term retained earnings gives the chapter a precise name for this part of the problem.*

**Capital market.** The market in which long-term money is raised by selling bonds and shares..

*The term capital market gives the chapter a precise name for this part of the problem.*

*The central question in debt and equity as sources is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of raising funds in the capital market before moving to application.*

**Initial public offering.** A company's first sale of shares to the public..

*The term initial public offering gives the chapter a precise name for this part of the problem.*

**Rights issue.** An offer of new shares to existing shareholders, in proportion to their holdings..

*The term rights issue gives the chapter a precise name for this part of the problem.*

**Underwriter.** An institution that promises to buy any shares of a new issue that the public does not..

*The term underwriter gives the chapter a precise name for this part of the problem.*

*The central question in raising funds in the capital market is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

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### 15.3 RAISING FUNDS IN THE CAPITAL MARKET

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The purpose here is to make raising funds in the capital market concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Initial public offering** means A company's first sale of shares to the public. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Rights issue** means An offer of new shares to existing shareholders, in proportion to their holdings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Underwriter** means An institution that promises to buy any shares of a new issue that the public does not. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For raising funds in the capital market, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### 15.4 THE COST OF CAPITAL

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The purpose here is to make the cost of capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

*This section builds the idea of the cost of capital before moving to application.*

In this chapter, **Cost of capital** means The return a firm must earn on its investments to satisfy those who supply its money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Weighted average cost of capital** means The average cost of a firm's different sources of money, each weighted by its share of the total. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the cost of capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to long-term financing and the capital market. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

**Cost of capital.** The return a firm must earn on its investments to satisfy those who supply its money..

*The term cost of capital gives the chapter a precise name for this part of the problem.*

**Weighted average cost of capital.** The average cost of a firm's different sources of money, each weighted by its share of the total..

*The term weighted average cost of capital gives the chapter a precise name for this part of the problem.*

*The central question in the cost of capital is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*



A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

*A short restatement keeps the financial meaning attached to the calculation.*

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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*A short restatement keeps the financial meaning attached to the calculation.*

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## SUMMARY

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The central purpose of Long-Term Financing and the Capital Market is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

*The chapter has connected its main terms, method, and business decision.*

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## THINK IT THROUGH

---

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

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## WHERE PEOPLE GO WRONG

---

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

*Most errors arise from mixing terms, timing, signs, or decision rules.*

before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*A historical case shows how the chapter's financial problem appeared in practice.*

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### IN HISTORY

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Dhaka Stock Exchange: its share-price boom of 2010 and the sharp fall of early 2011. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

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### REVIEW QUESTIONS

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15.T1 sort given facts into short, intermediate and long-term financing

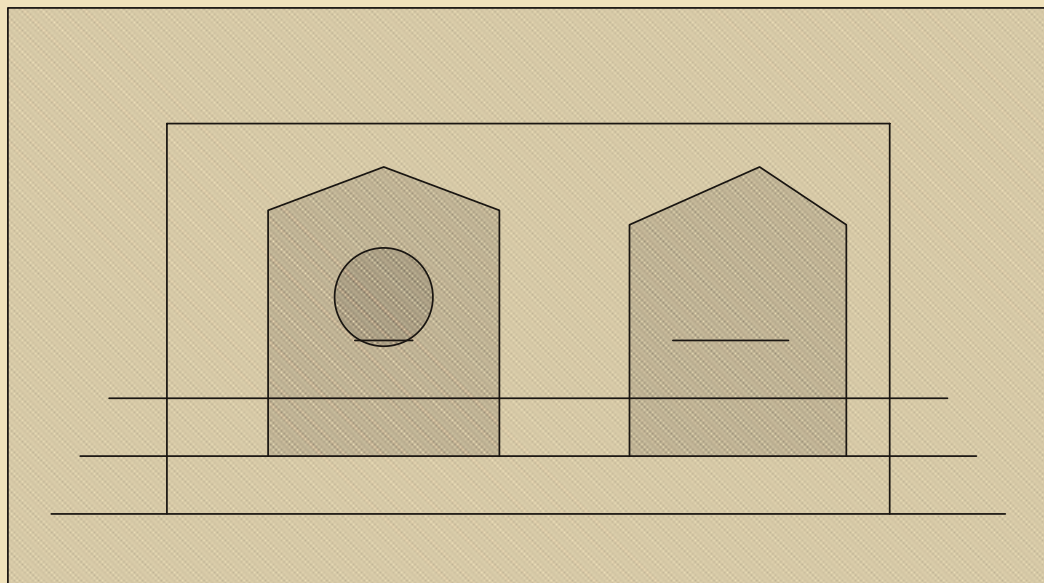
15.T2 describe the main sources of long-term money for a new company with their strengths and limits

15.T3 explain what retained earnings cost the firm

15.T4 compute the weighted average cost of capital for new figures

**PAUSE AND CHECK 15.P1** After the features of financing, state the rule in your own words.

**PAUSE AND CHECK 15.P2** After capital-market issues, state the rule in your own words.



*a carriage-hire yard with several carriages and a stable hand at work, 1880s*

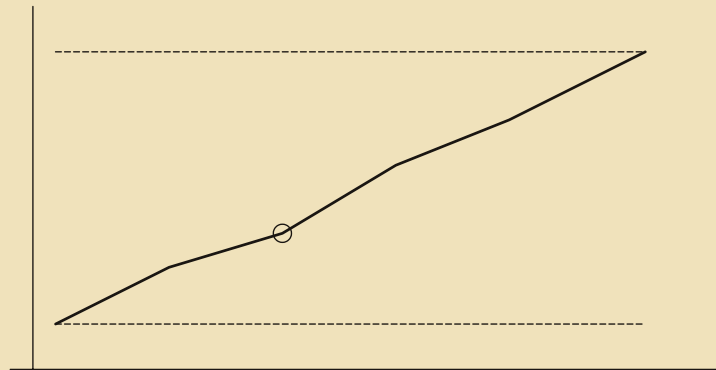
## CHAPTER XVI

# Leasing

*A practical question about leasing*



*This chapter introduces leasing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.*

FIG. 16.1. *Leasing figure 1*

*This section builds the idea of what leasing is before moving to application.*

**Lease.** A contract in which the owner of an asset lets another party use it for regular payments..

*The term lease gives the chapter a precise name for this part of the problem.*

**Lessor.** The owner who lets the asset out under a lease..

*The term lessor gives the chapter a precise name for this part of the problem.*

**Lessee.** The user who pays to use the asset under a lease..

*The term lessee gives the chapter a precise name for this part of the problem.*

*The central question in what leasing is is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

## 16.1 WHAT LEASING IS

The purpose here is to make what leasing is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Lease** means A contract in which the owner of an asset lets another party use it for regular payments. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Lessor** means The owner who lets the asset out under a lease. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Lessee** means The user who pays to use the asset under a lease. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what leasing is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a

larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 16.2 KINDS OF LEASE

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The purpose here is to make kinds of lease concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating lease** means A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial lease** means A long lease, not cancellable, that covers most of the asset's life and works like a loan. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sale-and-leaseback** means A deal in which a firm sells an asset it owns and leases it back from the buyer. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For kinds of lease, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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## 16.3 THE LEASE-OR-BUY DECISION

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The purpose here is to make the lease-or-buy decision concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the

*This section builds the idea of kinds of lease before moving to application.*

**Operating lease.** A short lease, cancellable, where the lessor keeps most of the risks of owning the asset..

*The term operating lease gives the chapter a precise name for this part of the problem.*

**Financial lease.** A long lease, not cancellable, that covers most of the asset's life and works like a loan..

*The term financial lease gives the chapter a precise name for this part of the problem.*

**Sale-and-leaseback.** A deal in which a firm sells an asset it owns and leases it back from the buyer..

*The term sale-and-leaseback gives the chapter a precise name for this part of the problem.*

*The central question in kinds of lease is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*This section builds the idea of the lease-or-buy decision before moving to application.*

**Tax shield.** The tax saved because a cost, such as interest or depreciation, reduces taxable income..

*The term tax shield gives the chapter a precise name for this part of the problem.*

**After-tax cash outflow.** A payment's cost after subtracting the tax it saves..

*The term after-tax cash outflow gives the chapter a precise name for this part of the problem.*

*The central question in the lease-or-buy decision is what changes when the underlying decision changes.*

*A small numerical example shows how the idea works before a more realistic case.*

*The business meaning matters as much as the arithmetic.*

*A complete worked example joins the chapter ideas in one decision.*

*The second pass checks both the arithmetic and the financial interpretation.*

*A comparison is useful only when both alternatives are measured on the same basis.*

chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Tax shield** means The tax saved because a cost, such as interest or depreciation, reduces taxable income. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **After-tax cash outflow** means A payment's cost after subtracting the tax it saves. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the lease-or-buy decision, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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### WORKED EXAMPLE

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Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to leasing. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value  $\times$  (1 + rate). Steps: Tk 100,000  $\times$  (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

*A short restatement keeps the financial meaning attached to the calculation.*

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*A short restatement keeps the financial meaning attached to the calculation.*

*The chapter has connected its main terms, method, and business decision.*

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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## SUMMARY

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The central purpose of Leasing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify

what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

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### THINK IT THROUGH

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

*Finance becomes stronger when a number is followed by a question about meaning.*

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### WHERE PEOPLE GO WRONG

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Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

*Most errors arise from mixing terms, timing, signs, or decision rules.*

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### IN HISTORY

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United States Leasing Corporation, founded in San Francisco in 1952, an early firm of the modern equipment-leasing industry. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

*A historical case shows how the chapter's financial problem appeared in practice.*

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### REVIEW QUESTIONS

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**16.T1** say what a lease is and compare operating and financial leases with a new asset

**16.T2** list the steps of the lease-or-buy comparison

**16.T3** compute the after-tax present value of a new lease and purchase

**16.T4** explain the tax shield of interest and depreciation

**PAUSE AND CHECK 16.P1** After kinds of lease, state the rule in your own words.

**PAUSE AND CHECK 16.P2** After after-tax outflows, state the rule in your own words.



APPENDIX A

# Maths You Will Use



*This section gives the arithmetic rule used in the book.*

*The tiny example shows the order of work.*

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---

## **A.1 A1.1 PERCENTAGES AND PERCENTAGE CHANGE**

---

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

---

## **A.2 A1.2 RATIOS AND AVERAGES**

---

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

---

## **A.3 A1.3 REARRANGING AN EQUATION AND ROUNDING**

---

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

---

## **A.4 A1.4 READING A GRAPH**

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Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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## GLOSSARY

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**Accounts payable.** Money a business owes its suppliers for goods already received. (*Chapter XIV*)

**Accounts receivable.** Money that customers owe a business for goods already delivered. (*Chapter XIV*)

**After-tax cash outflow.** A payment's cost after subtracting the tax it saves. (*Chapter XVI*)

**Agency cost.** A cost the principal bears to reduce the agency problem or from the loss it still causes. (*Chapter III*)

**Agency problem.** A conflict that arises when an agent's own interests differ from those of the principal. (*Chapter III*)

**Agent.** A person who acts for a principal and makes decisions for the principal. A hired shop manager is an agent. (*Chapter III*)

**Annuity.** A series of equal payments made at equal intervals for a fixed number of periods. (*Chapter VI*)

**Annuity due.** An annuity whose payments fall at the start of each period. (*Chapter VI*)

**Asset.** Something a business owns that has value, such as cash, stock or a van. (*Chapter II*)

**Balance sheet.** A statement listing what a business owns and what it owes on one date. (*Chapter II*)

**Beta.** A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. (*Chapter XI*)

**Bond.** A loan made to a company or government, repaid on a set date, with interest paid on the way. (*Chapter XII*)

**Break-even point.** The sales volume at which total revenue equals total cost, so profit is zero. (*Chapter X*)

**Business risk.** Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. (*Chapter IX*)

**Capital asset pricing model.** A formula for required return: risk-free rate plus beta times the market risk premium. (*Chapter XI*)

**Capital budgeting.** Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. (*Chapter VII*)

**Capital expenditure.** Spending on an asset that will help a business for more than one year. (*Chapter VII*)

**Capital gain.** The profit from selling an asset for more than its purchase price. (*Chapter XIII*)

**Capital market.** The market in which long-term money is raised by selling bonds and shares. (*Chapter XV*)

**Cash conversion cycle.** The operating cycle minus the days a business takes to pay its suppliers. (*Chapter XIV*)

**Cash discount.** A price cut a supplier gives for paying early, such as 2% off if paid within 10 days. (*Chapter XIV*)

**Cash flow.** Money that actually moves into or out of a business. Paying a supplier Tk 5,000 is a cash flow of minus Tk 5,000. (*Chapter I*)

**Cash inflow.** Money coming into a business or to an investor, shown as a positive number. (*Chapter V*)

**Cash outflow.** Money going out of a business or from an investor, shown as a negative number. (*Chapter V*)

**Coefficient of variation.** Standard deviation divided by expected value, giving risk per unit of expected return. (*Chapter IX*)

**Collateral.** An asset a lender may take if the borrower fails to repay. (*Chapter XV*)

**Common stock.** Shares that give owners a vote and a claim on profit after all other claims are met. (*Chapter XIII*)

**Company.** A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. (*Chapter I*)

**Compound interest.** Interest calculated on the principal sum and also on interest already earned. (*Chapter IV*)

**Compounding period.** The length of time after which interest is added to the balance, such as a year or six months. (*Chapter IV*)

**Constant growth model.** The dividend valuation model for dividends that grow at the same rate every year. (*Chapter XIII*)

**Contribution margin.** Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. (*Chapter X*)

**Controller.** The officer in charge of accounting records, reports and checking that money is used as planned. (*Chapter II*)

**Cost of capital.** The return a firm must earn on its investments to satisfy those who supply its money. (*Chapter XV*)

**Coupon rate.** The yearly interest of a bond, as a percentage of its face value. (*Chapter XII*)

**Covenant.** A condition in a loan agreement that the borrower promises to keep. (*Chapter XV*)

**Debenture.** A bond backed only by the issuer's general promise to pay, not by a particular asset. (*Chapter XV*)

**Depreciation.** The share of an asset's cost charged against income each year as the asset wears out. (*Chapter VII*)

**Discount bond.** A bond that sells for less than its face value. (*Chapter XII*)

**Discount rate.** The yearly percentage used in discounting; it states the return you could earn elsewhere. (*Chapter V*)

**Discounted payback period.** The payback time when each cash flow is first discounted to present value. (*Chapter VIII*)

**Discounting.** Finding present value by removing the interest a sum could earn before it arrives. (*Chapter V*)

**Diversification.** Spreading money across many investments so that one failure does less damage. (*Chapter XI*)

**Dividend.** A share of a company's profit paid out to its owners instead of being kept in the business. (*Chapter II*)

**Dividend decision.** A choice about how much profit to pay out to owners and how much to keep for the business. (*Chapter II*)

**Dividend valuation model.** A way of valuing a share as the present value of the dividends it is expected to pay. (*Chapter XIII*)

**Earnings per share.** Earnings left for owners of common stock divided by the number of shares. (*Chapter X*)

**EBIT.** Earnings before interest and tax: sales minus operating costs. (*Chapter X*)

**Efficient market.** A market in which prices already reflect the available information. (*Chapter XI*)

**Equity.** What is left for the owners when liabilities are subtracted from assets. (*Chapter II*)

**Expected value.** The average of possible results, each weighted by its probability. (*Chapter IX*)

**Face value.** The amount a bond repays at its end, for example Tk 1,000. (*Chapter XII*)

**Factor table.** A table of ready-made multipliers that saves repeating a long calculation. (*Chapter IV*)

**Finance.** The activity of finding money, using it, and judging whether the result was worth it. A shop owner who borrows to buy stock does finance. (*Chapter I*)

**Financial lease.** A long lease, not cancellable, that covers most of the asset's life and works like a loan. (*Chapter XVI*)

**Financial leverage.** The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). (*Chapter X*)

**Financial manager.** The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. (*Chapter II*)

**Financial planning.** Estimating the money a business will need and receive in future, and arranging how to meet the gap. (*Chapter II*)

**Financial risk.** Extra risk that comes from borrowing, because interest must be paid whatever the sales. (*Chapter IX*)

**Financing decision.** A choice about where the money comes from: owners' own funds, a loan, or new owners. (*Chapter II*)

**Fixed cost.** A cost that stays the same however many units are sold, such as shop rent. (*Chapter X*)

**Float.** Money tied up in payments that are sent but not yet usable, such as a cheque in the post. (*Chapter XIV*)

**Future value.** What a sum today will be worth at a later date when it earns interest. (*Chapter IV*)

**Incentive.** A reward, such as a bonus, that encourages a person to act in a chosen way. (*Chapter III*)

**Independent project.** A project that can be accepted or rejected without affecting any other project. (*Chapter VII*)

**Initial investment.** The cash flows at the start of a project to buy and prepare the asset. (*Chapter VII*)

**Initial public offering.** A company's first sale of shares to the public. (*Chapter XV*)

**Instalment.** One of a series of regular payments used to repay a loan or build a sum. (*Chapter VI*)

**Interest.** The payment a borrower makes to a lender for the use of money, or the reward a saver earns. (*Chapter IV*)

**Interest rate.** The interest for one period written as a percentage of the sum, for example 8% a year. (*Chapter IV*)

**Intermediate-term financing.** Money borrowed to be repaid in more than one year but not more than about five. (*Chapter XIV*)

**Internal rate of return.** The discount rate at which a project's net present value is exactly zero. (*Chapter VIII*)

**Inventory.** The goods a business holds for sale, such as the books on a shop's shelves. (*Chapter XIV*)

**Investment decision.** A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. (*Chapter II*)

**Lease.** A contract in which the owner of an asset lets another party use it for regular payments. (*Chapter XVI*)

**Lessee.** The user who pays to use the asset under a lease. (*Chapter XVI*)

**Lessor.** The owner who lets the asset out under a lease. (*Chapter XVI*)

**Leverage.** The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. (*Chapter X*)

**Liability.** Money a business owes to others, such as a bank loan or an unpaid supplier bill. (*Chapter II*)

**Long-term financing.** Money raised for more than about five years, or with no fixed repayment date. (*Chapter XIV*)

**Market return.** The average return of all investments in a market during a period. (*Chapter XI*)

**Matching principle.** Pay for short-lived needs with short-term money and for lasting needs with long-term money. (*Chapter XIV*)

**Maturity.** The date on which a bond or loan must be repaid. (*Chapter XII*)

**Mixed stream.** A series of cash flows of different sizes in different periods. (*Chapter V*)

**Mutually exclusive projects.** Projects of which only one can be chosen, because choosing one rules out the other. (*Chapter VII*)

**Net present value.** The present value of all a project's future cash flows minus its initial investment. (*Chapter VIII*)

**Net working capital.** Current assets minus current liabilities; what remains for trading after short debts are paid. (*Chapter XIV*)

**Operating cash flow.** The cash a project brings in each year after paying its cash costs and tax. (*Chapter VII*)

**Operating cycle.** The days from buying stock to collecting cash from its sale. (*Chapter XIV*)

**Operating efficiency.** How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. (*Chapter I*)

**Operating lease.** A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. (*Chapter XVI*)

**Operating leverage.** The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the

percentage change in EBIT divided by the percentage change in sales. (*Chapter X*)

**Opportunity cost.** What you give up by choosing one use of money instead of the best alternative. (*Chapter IV*)

**Ordinary annuity.** An annuity whose payments fall at the end of each period. (*Chapter VI*)

**Partnership.** A business owned by two or more people who agree to run it together and share its profit. (*Chapter I*)

**Payback period.** The time a project takes to return its initial investment from its cash flows. (*Chapter VIII*)

**Perpetuity.** A series of equal payments that continues for ever. (*Chapter XIII*)

**Portfolio.** A collection of investments held together by one investor. (*Chapter XI*)

**Preferred stock.** Shares that pay a fixed dividend, which must be paid before any dividend on common stock. (*Chapter X*)

**Premium bond.** A bond that sells for more than its face value. (*Chapter XII*)

**Present value.** What a sum due in the future is worth today. (*Chapter V*)

**Principal.** A person who hires another to act for them. A shop owner who hires a manager is a principal. (*Chapter III*)

**Principal sum.** The original amount lent, borrowed or deposited, before any interest is added. (*Chapter IV*)

**Probability distribution.** A list of possible results with the chance of each, the chances adding to one. (*Chapter IX*)

**Profit.** What a business earns from sales after subtracting the costs of earning them. Sales of Tk 100 and costs of Tk 80 leave profit of Tk 20. (*Chapter I*)

**Profit maximisation.** Aiming to make the largest profit possible, usually within one year. (*Chapter III*)

**Profitability index.** Present value of future cash flows divided by the initial investment. (*Chapter VIII*)

**Rate of return.** The return in one period divided by the sum invested, written as a percentage. (*Chapter IX*)

**Relevant cash flow.** A cash flow that changes because of the decision being studied. (*Chapter VII*)

**Required return.** The least return an investor will accept for taking a given risk. (*Chapter XI*)

**Residual claim.** The right to whatever is left after lenders and preferred owners have been paid. (*Chapter XIII*)

**Retained earnings.** Profit kept in the business instead of being paid out as dividends. (*Chapter XV*)

**Return.** The gain or loss from an investment, including income received and any change in value. (*Chapter IX*)

**Rights issue.** An offer of new shares to existing shareholders, in proportion to their holdings. (*Chapter XV*)

**Risk.** The chance that the actual result will differ from the result you expect. (*Chapter IX*)

- Risk premium.** The extra return demanded for taking risk instead of a risk-free investment. (*Chapter XI*)
- Risk-free rate.** The return on an investment with no risk of loss, such as a short-term government security. (*Chapter XI*)
- Sale-and-leaseback.** A deal in which a firm sells an asset it owns and leases it back from the buyer. (*Chapter XVI*)
- Security market line.** The graph of the capital asset pricing model, with beta on one axis and required return on the other. (*Chapter XI*)
- Semiannual interest.** Interest paid twice a year, each payment being half the yearly amount. (*Chapter XII*)
- Share price.** The price at which one share of a company can be bought or sold today. (*Chapter III*)
- Shareholder.** A person who owns part of a company by holding its shares. (*Chapter III*)
- Short-term financing.** Money borrowed to be repaid within one year. (*Chapter XIV*)
- Simple interest.** Interest calculated only on the principal sum, so it is the same every period. (*Chapter IV*)
- Sole proprietorship.** A business owned and run by one person, who keeps all profit and answers for all debts. (*Chapter I*)
- Stakeholder.** Anyone affected by a firm's actions, such as staff, customers, lenders and the community. (*Chapter III*)
- Standard deviation.** The square root of the variance; it shows how far results typically stray from the expected value. (*Chapter IX*)
- Sunk cost.** Money already spent that cannot be recovered whatever you decide now. (*Chapter VII*)
- Systematic risk.** Risk that affects all investments, such as a recession; it cannot be diversified away. (*Chapter XI*)
- Tax shield.** The tax saved because a cost, such as interest or depreciation, reduces taxable income. (*Chapter XVI*)
- Term loan.** A loan repaid in regular instalments over a fixed number of years, usually from a bank. (*Chapter XV*)
- Terminal cash flow.** The cash received or paid when a project ends, such as the sale of scrap. (*Chapter VII*)
- Time value of money.** The idea that a taka received today is worth more than a taka received later. (*Chapter IV*)
- Timeline.** A line marking each period, with the cash flow written at the point where it occurs. (*Chapter IV*)
- Total leverage.** The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. (*Chapter X*)
- Trade credit.** Credit from a supplier who lets a buyer pay some days after delivery. (*Chapter XIV*)
- Treasurer.** The officer in charge of handling cash, raising money and keeping relations with banks. (*Chapter II*)

**Underwriter.** An institution that promises to buy any shares of a new issue that the public does not. (*Chapter XV*)

**Unsystematic risk.** Risk that affects one firm or industry only; diversification can remove it. (*Chapter XI*)

**Valuation.** Finding what an asset is worth today by adding the present values of the cash it will bring. (*Chapter XII*)

**Variable cost.** A cost that rises and falls with the number of units sold, such as the price paid for each book. (*Chapter X*)

**Variance.** The probability-weighted average of the squared gaps between each result and the expected value. (*Chapter IX*)

**Wealth maximisation.** Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. (*Chapter III*)

**Weighted average cost of capital.** The average cost of a firm's different sources of money, each weighted by its share of the total. (*Chapter XV*)

**Working capital.** The money a business has tied up in everyday trading: cash, stock and money owed to it. (*Chapter XIV*)

**Yield to maturity.** The yearly return an investor earns by buying a bond at today's price and holding it to maturity. (*Chapter XII*)



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## ANSWERS TO QUESTIONS

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### CHAPTER I · WHAT FINANCE IS

**01.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**01.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**01.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**01.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**01.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

### CHAPTER II · THE FINANCIAL MANAGER'S DECISIONS

**02.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**02.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**02.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**02.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**02.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

### CHAPTER III · THE GOAL OF THE FIRM AND AGENCY THEORY

**03.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**03.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**03.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**03.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**03.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER IV · INTEREST AND FUTURE VALUE

**04.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**04.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**04.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**04.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**04.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER V · PRESENT VALUE AND DISCOUNTING

**05.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**05.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**05.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**05.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**05.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER VI · ANNUITIES

**06.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

reasoning should use the facts given in the question and keep timing and units consistent.

**06.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**06.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**06.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**06.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**06.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER VII · CAPITAL BUDGETING: PROCESS AND CASH FLOWS

**07.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**07.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**07.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**07.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**07.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**07.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER VIII · EVALUATING PROJECTS: PAYBACK AND NET PRESENT VALUE

**08.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**08.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

reasoning should use the facts given in the question and keep timing and units consistent.

**08.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**08.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**08.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER IX · RISK AND RETURN

**09.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**09.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**09.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**09.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**09.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER X · LEVERAGE

**10.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**10.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**10.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**10.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**10.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**CHAPTER XI · REQUIRED RETURN, CAPM AND MARKET EFFICIENCY**

**11.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**11.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**11.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**11.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**11.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**11.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**CHAPTER XII · VALUATION OF BONDS**

**12.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**12.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**12.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**12.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**12.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**12.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**CHAPTER XIII · VALUATION OF SHARES**

**13.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

reasoning should use the facts given in the question and keep timing and units consistent.

**13.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**13.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**13.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**13.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER XIV · WORKING CAPITAL AND SHORT-TERM FINANCING

**14.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**14.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**14.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**14.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**14.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**14.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER XV · LONG-TERM FINANCING AND THE CAPITAL MARKET

**15.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**15.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**15.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

reasoning should use the facts given in the question and keep timing and units consistent.

**15.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**15.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**15.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

#### CHAPTER XVI · LEASING

**16.t1** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**16.t2** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**16.t3** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**16.t4** A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

**16.p1** State the rule introduced at that point, then give one sentence explaining why it fits the problem.

**16.p2** State the rule introduced at that point, then give one sentence explaining why it fits the problem.



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